AIR FORCE TACTICS, TECHNIQUES, AND PROCEDURES 3-32.33, VOLUME 1

14 APRIL 2026

EXPEDIENT HYGIENE, WATER, AND WASTE METHODS



DEPARTMENT OF THE AIR FORCE

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BY ORDER OF THE SECRETARY OF THE AIR FORCE

AIR FORCE TACTICS, TECHNIQUES, AND PROCEDURES 3-32.33V1

14 April 2026

Tactical Doctrine

EXPEDIENT HYGIENE, WATER, AND WASTE METHODS

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PURPOSE: This Air Force Tactics, Techniques and Procedures (AFTTP) supports Air Force Instruction (AFI) 10-210, *Prime Base Engineer Emergency Force (BEEF) Program*, and AFI 10-209, *RED HORSE Program*. This publication applies to the Regular Air Force, the Air Force Reserve, and the Air National Guard. This publication does not apply to the United States Space Force. This AFTTP provides civil engineer (CE) craftsmen with ideas, options, and methods for establishing temporary or expedient hygiene, water, and waste facilities at austere, remote, or dispersed contingency locations when standard equipment and materials are not available and when necessary for minimum-essential restoration of services. Refer recommended changes and questions about this publication to the Office of Primary Responsibility using the Department of the Air Force (DAF) Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command and Major Command publications/forms managers. Ensure all records generated as a result of processes prescribed in this publication adhere to AFI 33-322, *Records Management and Information Governance Program* and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the DAF.

SUMMARY OF CHANGES

This document has been revised and should be completely reviewed. In addition to updating terminology, references, and figures, other changes include the removal of redundant information from other AFTTP 3-32-series and AFMAN 32-series civil engineer publications, including data pertaining to non-expedient septic systems; site evaluations and soil testing; water wells and well drilling; packaged and field-erected incinerators, solid waste recycling and reuse, and sanitary and engineered landfills.

APPLICATION: This AFTTP is nondirective and does not replace policy documents, technical orders, instructions, manuals, or any other mandatory guidance relevant to techniques and procedures in this publication. The expedient methods presented may be applicable during wartime or overseas contingency operations and force beddown in remote or austere locations with shortages of hygiene, water, and waste facilities or standard construction materials. Some expedient methods may be an option when immediate action is necessary to restore service following a hostile attack or disaster.

SCOPE: The information in this publication does not supersede any legal requirements (domestic US law for operations in the US and for operations outside the US: US law with extraterritorial application, applicable international agreement requirement, or combatant command directive) that specify how tasks should be performed. In accordance with AFI 10-210, Prime BEEF contingency activities comply with environmental provisions of all applicable international agreements and requirements in Department of Defense Instruction (DoDI) 4715.22; *Environmental Management Policy for Contingency Locations*, DoDI 4715.19, *Use of Open Air Burn Pits in Contingency Operations*; applicable combatant command directives and operation plans, operation orders, fragmentary orders, or similar operational directives for specific contingency operations (e.g., Annex L, Environmental Considerations to the Combatant Command Operation Plan); and relevant Lead Service policy (e.g., location specific environmental management plans).

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Chapter 1

INTRODUCTION

1.1. Overview. This AFTTP addresses field expedient and improvised methods for sanitation and hygiene facilities, surface water source development, wastewater, and solid waste disposal at austere, remote, or dispersed contingency locations when time and materials are limited, or when conventional methods are impractical to employ. It does not address expedient repairs to existing facilities or the installation of standard contingency hygiene, water, and waste systems such as Basic Expeditionary Airfield Resources (BEAR). For the purpose of this document, expedient is "a means devised or employed in a time and place where prompt action is essential." Expedient engineering does not preclude using normal engineer practices where time and materials are available. This publication contains illustrations of basic material applications and field-expedient construction techniques and procedures. Any data and definitions presented are general in nature and serve as field guides. See **Attachment 1** for a glossary of references and supporting information.

1.2. Environmental Considerations. The health and safety of personnel are critical in any military operation. Safe food and water, and a means of properly disposing of waste are essential. The intention of environmental goals during contingencies is to minimize risks to human health and safety and prevent unnecessary damage to the environment while maximizing the natural resources available to support readiness and operational effectiveness. More information on integrating environmentally responsible practices during contingency operations can be found in DoDI 4715.22 and Air Force Handbook (AFH) 10-222V4, *Environmental Considerations for Overseas Contingency Operations*.

1.3. General Safety Practices. Every civil engineer job or operation has its own particular safety hazards, and everyone involved should be alert to prevent injury or illness. This is especially critical during force beddown, expedient construction activities at austere locations, and base recovery after an attack or disaster. Keep safety in mind when using expedient methods and materials described in this publication. Adhere to technical data warnings and cautions and wear personal

protective equipment (PPE) and individual protective equipment for the working environment. **Note:** Department of the Air Force Manual (DAFMAN) 91-203, *Department of the Air Force Occupational Safety, Fire, and Health Standards*, lists PPE for specific hazards and activities.

Chapter 2

EXPEDIENT HYGIENE AND SANITATION

2.1. General Information. During contingency deployments and beddown operations at remote, austere, and dispersed locations, sanitation facilities for disposal of human waste and basic personal hygiene are essential to maintaining good health. Generally, forces at contingency locations use prepackaged deployable facilities (i.e., BEAR assets) or contracted portable facilities when sanitary systems are inadequate or unavailable. Improvised or field expedient methods that provide crude, temporary amenities may be necessary in remote areas until deployable or contracted facilities arrive, or after an attack or disaster, or when existing sanitary facilities are being rehabilitated. **Note:** Local, state, federal, and host-nation regulations or laws may prohibit burying human waste. Consult theater waste management guidance for requirements.

2.2. Overview. This chapter presents basic methods for creating expedient, rudimentary facilities to support initial or emergency sanitation and hygiene requirements at contingency locations. These methods are not considered long-term solutions. Facilities addressed in this chapter include field expedient latrines, urinals, hand-washing stations, and shower-shave stations. For additional information on sanitation and waste disposal procedures at contingency locations, refer to DoDI 4715.19, AFH 10-222V4, AFTTP 3-32.34V1, *Civil Engineer Bare Base Development*, and AFPAM 10-219V5, *Bare Base Conceptual Planning*.

2.3. Field-Expedient Latrines. The field expedient latrines selected for a given situation depend on a number of factors–the number of personnel, the extent of the emergency or duration of stay at the site, and geological and climatic conditions. In all situations, sanitation and maintenance will be critical to prevent disease transmission to and from personnel. Always consult the Preventive Medicine team when planning and siting latrine facilities. They can assist in determining the right type, number, size, and location of latrines and sanitation facilities. Latrine locations are usually a compromise between separation requirements from dining facilities and water sources, and convenience for personnel. Multiple latrine sites will be necessary for larger contingency

locations. The field expedient latrines listed in **Table 2.1** are basic versions of either pit or aboveground latrines, including drum-type latrines. **Note:** Burn-out latrines and pail latrines if disposed of by open-air burns, for the collection of human waste (solid/liquid excretory waste and toilet paper), will adhere to DoDI 4715.19 as described in **paragraph 5.7** below for solid waste.

Table 2.1. Pit and Aboveground Field-Expedient Latrines.

Expedient Latrines
Cross Tree Latrines
Straddle Trench Latrines
Deep Pit Latrines
Bored Hole Latrines
Burn-Out Latrines
Cat Hole Latrine
Mound Latrine
Pail Latrines
Urine Soakage Pits

2.4. Pit Latrine. The pit latrine addressed in this chapter includes cross tree, straddle tree, deep pit, bore hole, and cat hole latrines. All versions of the pit latrine involve digging a pit in the ground. Pit latrines should be located where the groundwater table is deep enough to prevent groundwater contamination or water standing in the latrine pit. Its use is limited to areas that are free of impervious rock formations near the surface. See **paragraph 2.5** for alternatives to pit latrines where a high groundwater table or a rock formation near the surface prevents digging a pit of adequate depth.

2.4.1. Basic field requirements. The following general rules apply to constructing all types of pit latrines. To ensure that food and water are protected from

contamination, latrines should be at least 100 yards from the dining facility and 100 feet from the nearest water source. Latrines should not be dug below the groundwater table or where they may drain into a water source. The groundwater table can be determined from information given by local inhabitants or by excavating down to the groundwater table. Latrines are usually located at least 100 feet from the end of billeting areas but within a reasonable distance for easy access. They should be lighted at night if the military situation allows. If lights cannot be used, tie pieces of cord or tape to trees or stakes as guides to the latrines. **Note:** Latrines must be policed daily. Specific unit personnel are to be assigned the responsibility of ensuring that the latrines are always properly maintained.

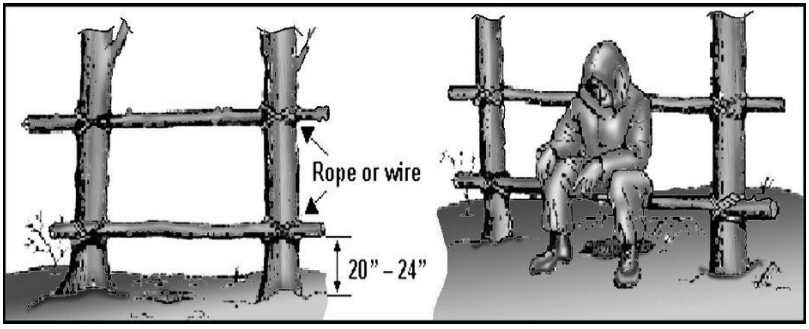
2.4.1.1. Drainage and fly control. Place a canvas or brush screen around each latrine or enclose within a tent. If possible, heat the tent in cold climates. Dig a drainage ditch adjacent to latrine facilities to prevent water from pooling around or flowing into the latrine. To control flies and other disease-transmitting insects, maintain a good fly control program in the area. **Note:** Do not spray pit contents. Comply with manufacturer's directions when applying and using insecticides.

2.4.1.2. Closing procedures. Close a latrine pit when it is filled to within 1 foot of the surface or when it is abandoned. Remove the latrine box and close as follows:

- Fill pit to the surface with successive, 3-inch layers of earth and pack each layer down.
- Place a l-foot mound of dirt over the length of the pit to prevent fly pupa from escaping the closed latrine.
- Install a rectangular sign on top of the mound to indicate the type of pit and the date closed; include the unit designation in non-operational areas.

2.4.2. Cross tree latrine. The cross tree latrine (**Figure 2.1**) is a basic, improvised latrine that will usually serve the needs of 6 to 8 personnel. Situate the latrine downwind from the beddown area, but not so far from shelters as to encourage individuals to break sanitary discipline. In areas where it is difficult to dig a pit, use ration boxes or similar material should collect waste. Improvise a windbreak using boughs, tarpaulins, ponchos, or a snow wall to provide both privacy and protect the latrine from wind.

Figure 2.1. Cross Tree Latrine.



2.4.3. Straddle trench latrine. The most common type of latrine for temporary (one to three days) beddowns is the straddle trench latrine (sometimes referred to as straddle pit latrine). The straddle trench latrine (**Figure 2.2**) is usually dug 1 foot wide, 2-1/2 feet deep, and 4 feet long. In this configuration, each trench will accommodate two people at the same time. Provide straddle trenches to serve at least 4 percent of the male population and 6 percent of the female population. For example, a unit of 100 men and 100 women needs at least four latrines for men and six for women. Place trenches at least 2 feet apart. There are no seats with this type of latrine; however, placing boards along both sides of the trench provides better footing. Place toilet paper on a suitable holder and protect it from weather with a tin can or other covering. Remove earth and pile it at the end of the trench so that each individual can properly cover his excreta and toilet paper. **Figure 2.3** depicts straddle trenches with individual privacy screens and windbreaks.

Figure 2.2. Straddle Trench Latrine.

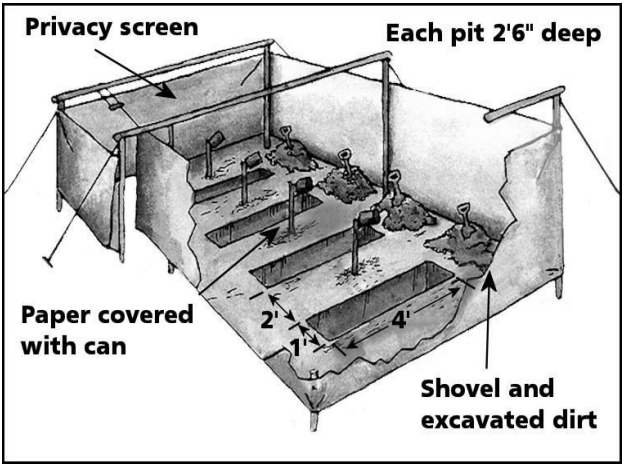


Figure 2.3. Straddle Trench Latrines with Individual Privacy Screen.

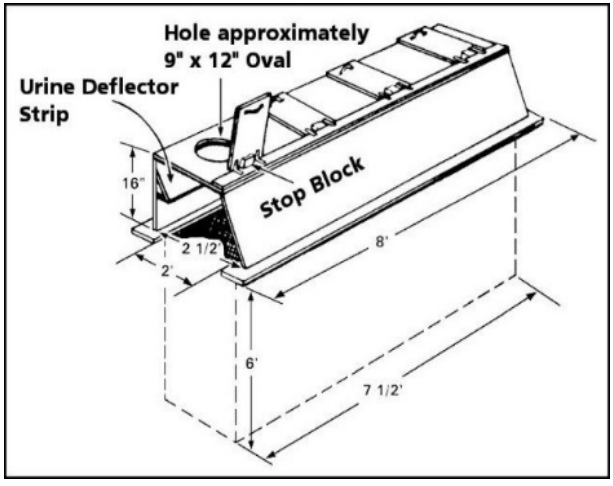


2.4.4. Deep pit latrine. The deep pit latrine is used for longer periods and often in built-up areas. As shown in **Figure 2.4**, it is used with a latrine box. Although

two-seat latrine boxes are sometimes used, the standard latrine box has four seats and is 8 feet long and 2-1/2 feet wide at the base. Dig the pit about 2 feet wide and 7-1/2 feet long. This gives the latrine box 3 inches of support on all sides. The pit depth depends on the latrine's period of use. As a rough guide, allow 1 foot of depth for each week of estimated use, plus 1 foot for the dirt cover when closed.

2.4.4.1. Rock or high groundwater levels often limit the depth of the pit, but the pit should be no deeper than 6 feet. For some soils, support bracing may help to prevent the sides from collapsing; use planking or similar material. Pack earth tightly around bottom edges of the box to seal openings where flies might enter. **Note:** If the ground is too hard for digging or the water table is too high, consider using a **pail latrine** or a **burnout latrine** in place of the deep pit latrine. See each alternative for additional information.

Figure 2.4. Four-Seat Deep Pit Latrine.



2.4.4.2. Cover holes with fly-proof, self-closing lids. Fly proof cracks with strips of wood or tin. Place a metal deflector (e.g. a flattened can) inside the front of the box to prevent urine from soaking into the wood.

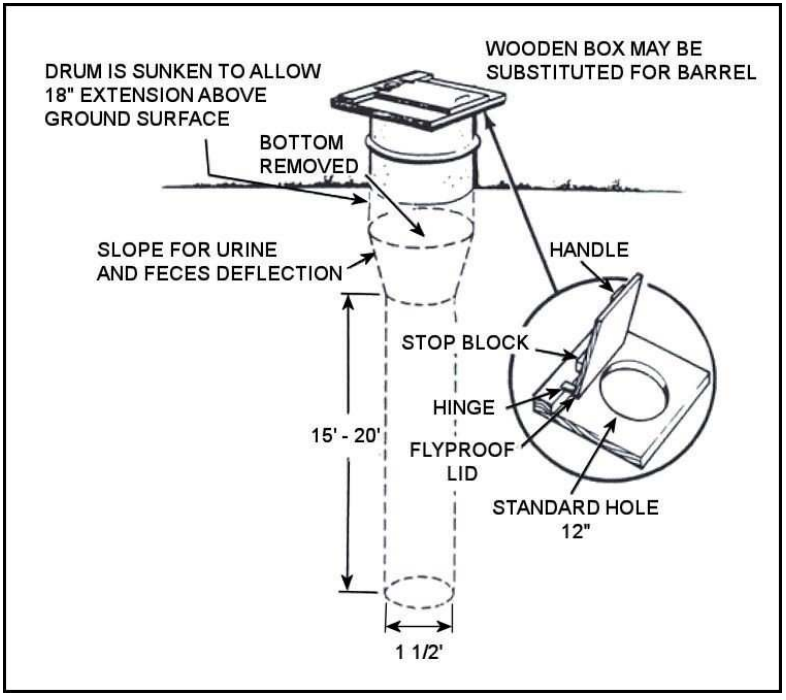
2.4.4.3. To prevent flies from breeding and to reduce odors, keep the latrine box clean, seat lids closed, and cracks sealed. Maintain a good fly control program in the area. Applying lime or burning pit contents does not effectively control flies or odor. Scrub the box and latrine seats daily with soap and water daily.

2.4.4.4. Create a portable version of the deep pit latrine by building the latrine box inside a portable structure that offers limited privacy and protection from the elements. If prepackaged deployable sanitation assets experience delays, this more substantial portable latrine may serve large groups better than some of the other expedient options discussed in this chapter.

2.4.4.5. Although portable latrines vary in design and construction, most use the same basic principles of mobility, adequate ventilation, proper sanitation, and reasonable privacy. Once the pit reaches capacity, simply relocate the latrine to another spot (usually in the same vicinity), then fill and close the old pit as discussed in **paragraph 2.4.1.2**.

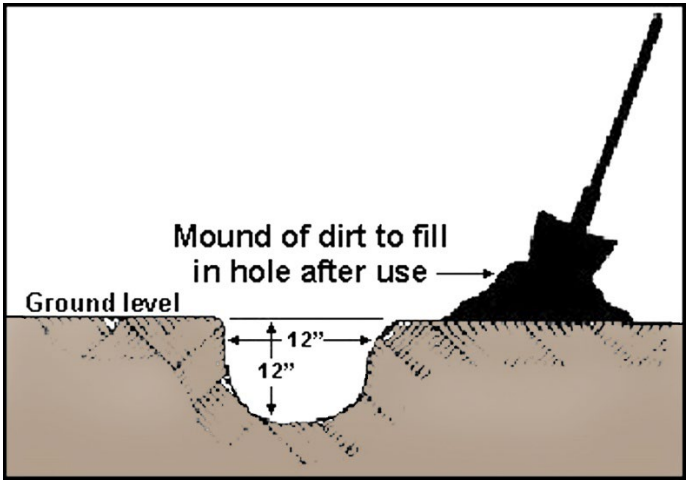
2.4.5. Bored hole latrine. A bored hole latrine consists of a hole that is about 18 inches in diameter and 15 to 20 feet deep (**Figure 2.5**). The actual diameter is not critical, so make it as large as available augers permit. Remove both ends of a metal drum and sink it into the ground to serve as a box. Make a fly-proof seat cover with a self-closing lid to fit the top of the drum. If a drum is not available, construct a fly-proof, 18-inch-high wooden box. A bored hole latrine is better suited for small units.

Figure 2.5. Bored Hole Latrine.



2.4.6. Cat hole latrine. The simplest of all pit latrines and field human waste disposal devices is the cat hole latrine (see **Figure 2.6**). Usually, this type of latrine is used by individuals on the move or when on patrol (if individual waste collection bags are not available), and in situations where onsite latrine facilities are not available, and digging individual cat holes is the last option. In any case, personnel should try to dispose of the waste immediately. As indicated in the drawing, a cat hole latrine should be dug at least 1 foot wide and 1 foot deep. After use, replace and repack the removed soil.

Figure 2.6. Cat-Hole Latrine.

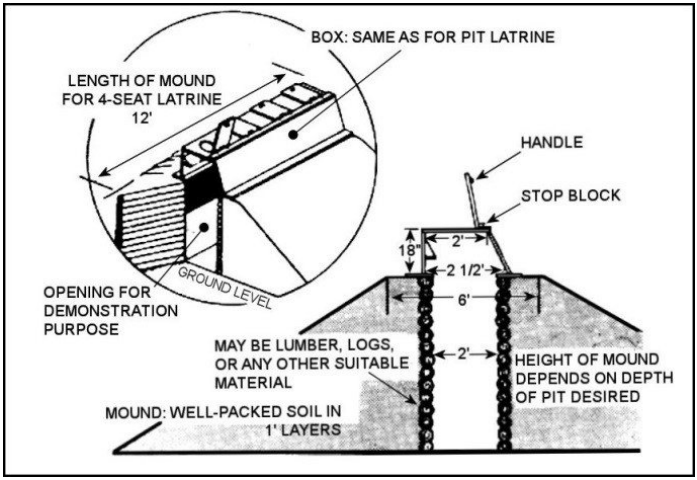


2.5. Aboveground and Drum Type Latrines. Aboveground and drum-type latrines are normally used when the groundwater table is high and where groundwater contamination or standing water is probable in the latrine pit. Aboveground and drum-type latrines may also be ideal when an impermeable rock formation is close to the surface, preventing adequate dispersal of liquids.

2.5.1. Mound latrine. A dirt mound makes it possible to build a deep-pit latrine without the pit extending into water or rock. Construct a mound of earth that is at least 6 feet wide and 12 feet long. It should be able to support a 4-hole latrine box. The mound should be high enough to meet the pit's depth requirement. Allow 1-foot from the base of the pit to the water or rock level. Break up or plow the area where latrine will be located to aid in seepage of liquids from the pit. If timber is available, build a crib of desired height to enclose the pit and support the latrine box. Build the mound and compact it in successive l-foot layers to the top of the crib as shown in **Figure 2.7**. Roughen the surface of each layer before adding the next. If timber for a crib is unavailable, construct the mound to the desired height in l-foot layers, as described and dig the pit into the mound. If necessary, brace

the walls with wood, sandbags, or other material to prevent them from collapsing. Fly proof and enclose a mound latrine as described in **paragraph 2.4.1.1**. **Note:** The size of the mound base depends on the type of soil in the area. Make the mound larger if the slope is steep. In addition, it may be necessary to build steps up a steep slope.

Figure 2.7. Mound Latrine.



2.5.2. Burn out latrine. Often, burn out latrines (**Figure 2.8**) are the preferred method for improvised latrines. These latrines are particularly suitable for jungle areas with high groundwater tables and in desert regions where an impervious rock layer is a short distance under the sand. Ensure the latrine location is downwind of the base camp. A unit of 100 men and 100 women will need at least eight men's latrines and eight women's latrines. As shown in **Figure 2.9**, consolidating burn out latrines within a multiple-station latrine facility with floors, cover, and partial walls provides semi-privacy and ease of maintenance.

Figure 2.8. Burn Out Latrine Details.



Figure 2.9. Multiple-Station Burn Out Latrines.



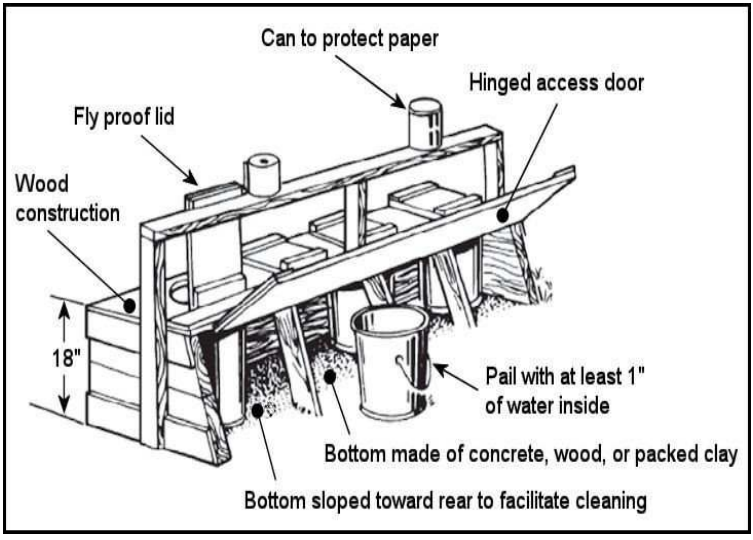
2.5.2.1. The simplest method to create a basic burn out latrine is to place a 55-gallon drum in the ground and leave enough of the drum above the ground for a

comfortable sitting height. Then place a wooden seat with a fly-proof, self-closing lid on top of the drum. However, for easier handling, cut the drum in half, making two latrines of less capacity. Weld handles to the sides of the drum. The handles will allow two people to carry the drums to a suitable location to burn the contents. If possible, have two sets of drums so while one set is in use, the other set is being burned out. Encourage male personnel to urinate in a urine disposal facility (see **paragraph 2.6**) rather than a burn out latrine because more fuel is required to clean a latrine with a liquid content.

2.5.2.2. Burn the contents of each drum daily by adding sufficient fuel to incinerate the fecal matter. Do not use highly volatile fuel because of its explosive nature. A mixture of 1 quart of gasoline to 5 quarts of diesel oil is effective; nevertheless, use it with caution. Burn the contents again if they do not become dry and odorless in one effort. Bury the residual ash once the burning process is complete.

2.5.3. Pail latrine. Similar to burn-out latrines, consider use of pail latrine facilities when conditions are such that a pit latrine is not feasible (e.g., populated areas, rocky soil, marshes). Construct a standard latrine box according to **paragraph 2.4.4**. Add a floor and place a pail under each seat. Place hinged doors on the rear of the box (**Figure 2.10**) to facilitate removal of pails when emptying. If the box is located within an enclosure, position the box to form a part of the outer wall and ensure the rear of the box opens directly to the outside of the building. The box should be fly proof, and seats and rear doors should be self-closing. Construct the floor of the box with an impervious material (concrete if possible) and allow sufficient slope toward the rear to allow washing water to drain quickly. Clean pails at least once daily and bury, burn, or dispose of the contents by another sanitary method. Plastic liners for the pails reduce the risk of accidental spillage. Tie the filled bags at the top before disposal.

Figure 2.10. Pail Latrines.



2.5.4. Makeshift and portable waste devices. A lack of resources in austere locations could make it difficult to build the aforementioned field latrines. In such situations, makeshift or portable waste devices described below may serve as a temporary solution. If available, consider using individual waste aggregation bags for solid human waste in the very early stages, since the bags are biodegradable, spill-proof, and can be disposed of in any landfill.

2.5.4.1. During an emergency or during field activities where digging a hole is not an option, personnel can make an expedient toilet using a common water bucket. Simply line a bucket with a garbage bag and make a toilet seat out of two boards placed parallel to each other across the bucket. After each use, pour a disinfectant such as bleach (1 part liquid chlorine bleach to 10 parts water) into the garbage bag to help avoid infection and stop the spread of disease. Cover the bucket tightly when it is not in use. Bury the garbage and human waste in a pit two to three feet deep and at least 50 feet downhill or away from any well, spring, or water supply.

2.5.4.2. For short periods or special operations in remote and austere locations when no other facilities are available, personnel may have access to portable human waste devices similar to those listed in **Table 2.2**.

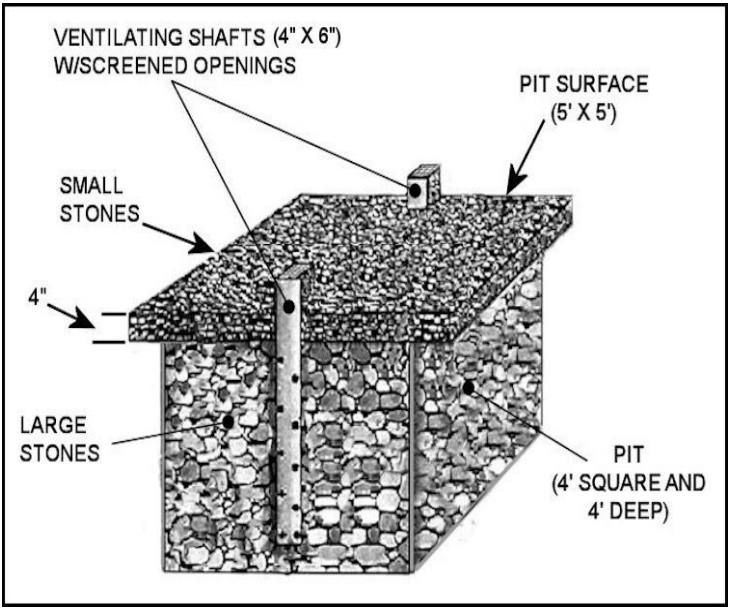
Table 2.2. Portable and Disposable Human Waste Devices.

Device	National Stock Number
Urinal, portable, female	8530-01-470-2805
Urinal bag, disposable	4510-01-379-0177
Portable latrine bag (for solid waste)	4510-01-379-1341
Portable restroom kit (for liquid and solid wastes)	4510-01-379-0190
Portable commode	4510-01-423-0100 / 4510-01-382-4289

2.6. Urinals. In permanent and semi-permanent camps, urine disposal facilities usually connect to the sewer system. However, in a field environment, separate devices for urine disposal may be necessary. Collocate urinal facilities in or near the male latrines to minimize soiling latrine seats and liquid content in burn-out latrines. At least one urine disposal facility is required for each male latrine or per 100 personnel. The following paragraphs describe examples of the more common urine disposal expedient field devices and their applicable maintenance procedures.

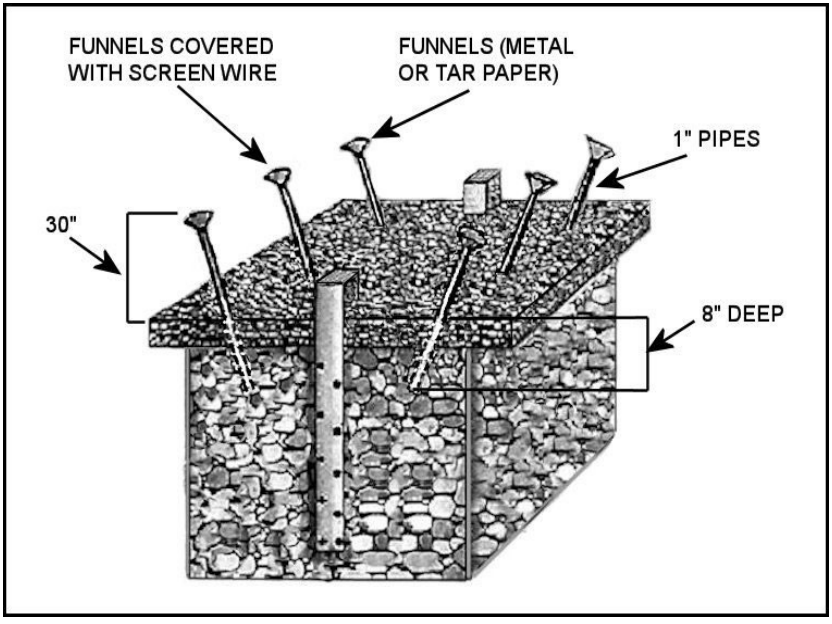
2.6.1. Urine soakage pit. One of the best expedient devices for urine disposal is a urine soakage pit. Begin by digging a pit 4 feet square and 4 feet deep, fill it with an aggregate material and lay a border along each edge so that each side of the soakage pit's surface is 5 feet long. Ideally, the border should be 6 inches wide, 4 inches deep, and composed of small stones. An optional feature is ventilating shafts as shown in **Figure 2.11**. Finish the facility by adding either a pipe or a trough urinal as described in the following paragraphs.

Figure 2.11. Urine Soakage Pit.



2.6.2. Pipe urinal soakage pit. Pipe urinals should be at least 1 inch in diameter and installed at an angle near each corner of the pit or, if needed, on the sides halfway between the corners (**Figure 2.12**). The pipes should extend at least 8 inches below the surface of the pit. Place a funnel made of tarpaper, sheet metal, or similar material in the top of each pipe. The upper rim of the funnel should extend about 30 inches above the pit's surface. Place a suitable screen in the funnel to prevent foreign objects from clogging the pipe.

Figure 2.12. Pipe Urinal Soakage Pit.



2.6.3. Trough urinal soakage pit. If materials are available, another option is to build a trough urinal as shown in **Figure 2.13**. The trough is U- or V-shaped and made of sheet metal or wood. If constructed of wood, line the trough with heavy tarpaper. Cut the trough legs slightly shorter on the end nearest the pit so the urine flows into the soakage pit. Illustrated in **Figure 2.14** is a variation of the expedient urinal trough. When building this type of trough, the four troughs should be no more than 4-1/2 feet long when used with the soakage pit addressed in **paragraph 2.6.1**. Each trough should slope slightly toward one corner where a pipe carries the urine to the soakage pit.

Figure 2.13. Trough Urinal Soakage Pit.

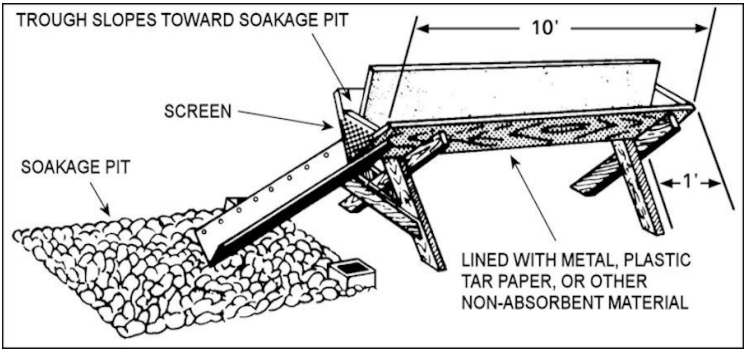
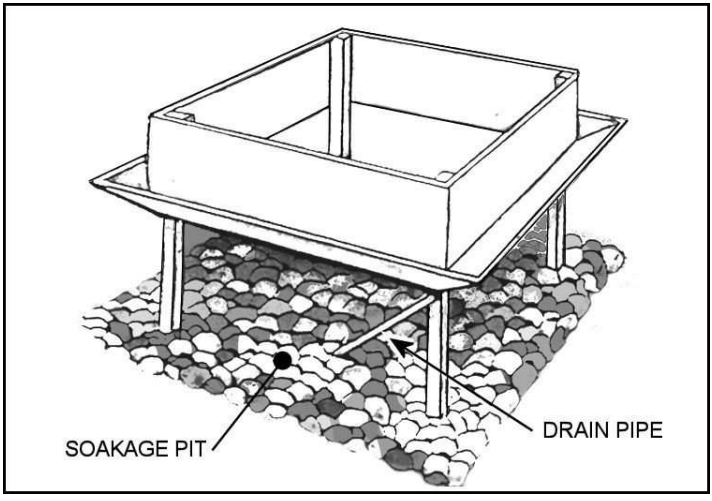
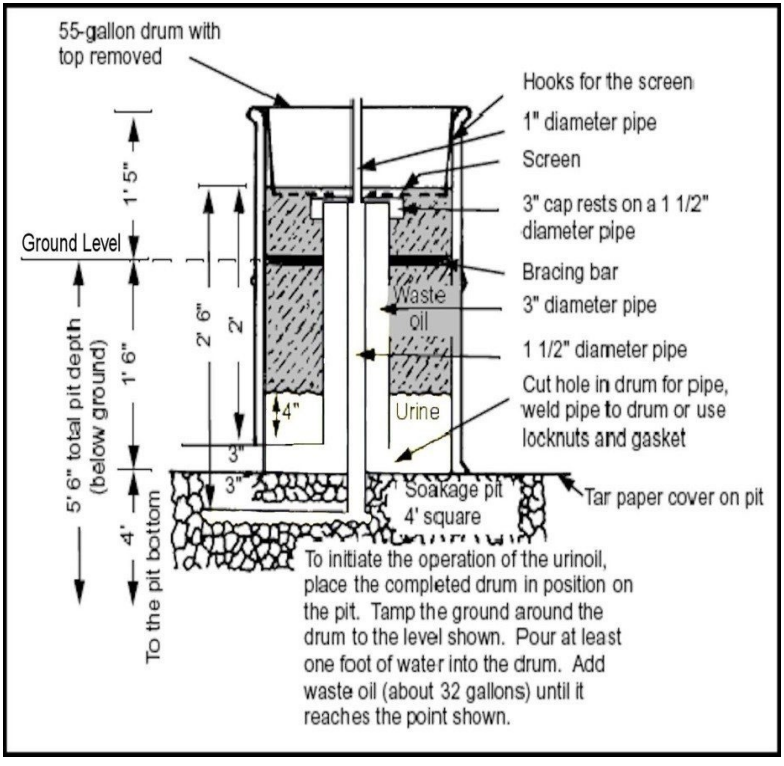


Figure 2.14. Trough Urinal Soakage Pit Variation.



2.6.4. Urinoil. The urinoil is an expedient urine disposal system appropriate for longer deployments. In its simplest form, the urinoil is a 55-gallon drum containing oil and sitting on top of a recessed soakage pit (**Figure 2.15**). Vegetable oil is preferred, but waste petroleum, oil, and lubricants (POL) may be used. The urinoil operational method is as follows:

Figure 2.15. Basic Urinoil Details.



2.6.4.1. Urine voided through the screen immediately sinks through the oil to the bottom of the drum. The action of the urinal is somewhat like that of a barometer. Adding more urine results in the oil level rising in the 3-inch pipe. This continues until it reaches the 1-1/2-inch notch on the overflow pipe in the center of the drum. Atmospheric pressure and the weight of the oil cause the urine to overflow until reestablishing equilibrium in the drum. The oil also acts as an effective seal against odors and flies.

2.6.4.2. To initiate the operation of the urinoil, place the prepared drum in position on the soakage pit and tamp the ground around the drum to the level shown in the drawing. Next, pour at least one foot of water into the drum. Finally, add oil (about 32 gallons) until the oil reaches the point indicated in the graphic. The urinoil will remain functional as long as the soakage pit will accept urine.

2.6.5. Maintenance of urine disposal facilities. Daily inspections are required at urine disposal facilities. Easily lift the screen with attached hooks to remove debris. **Table 2.3** lists additional steps for maintenance of urinals and soakage pits.

Table 2.3. Urinal Facilities Maintenance.

Urinal/Soakage Pit Maintenance	
Step 1	For proper operation and sanitary purposes, individuals should urinate in the trough or the pipe (whichever is available), not directly on the pit's surface.
Step 2	Wash funnels and troughs with soap and water daily.
Step 3	Replace funnels when necessary.
Step 4	Prevent oil or grease from getting into the pit to prevent clogging. Oil leeching through the pit may also contaminate groundwater.

Step 5	If the latrine is located some distance from sleeping areas, place a large can or pail at a convenient location for use as a urinal at night. Empty the can into the trough, pipe, or soakage trench every morning, and wash the pail with soap and water before reusing it.
Step 6	When a urine soakage pit is abandoned or becomes clogged, spray it with residual insecticide. Mound it over with an l-foot covering of compacted earth and place a rectangular sign on the mound indicating the type of pit and date of closure.

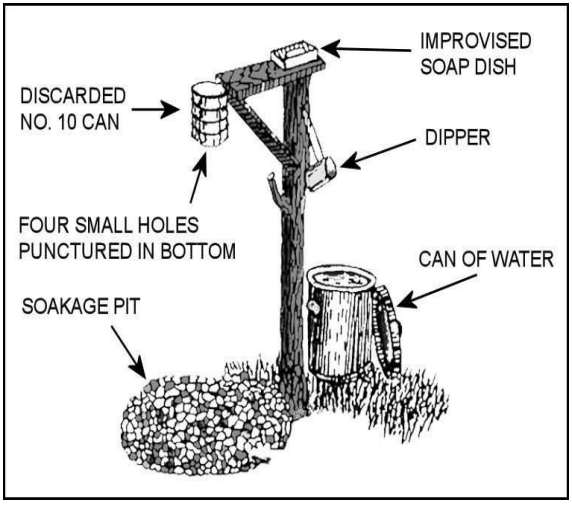
2.7. Expedient Wash and Shower Stations. Just as field expedient latrines, field expedient wash and shower stations are crude but functional devices used when other more appropriate and desirable facilities are not accessible. Hand washing stations and showers are generally located in areas convenient for base residents. Typically, engineer personnel are responsible for the construction and upkeep of expedient wash and shower stations to include disposal of the resulting wastewater (gray water). While it is preferable to dispose of wastewater into a base sanitary sewer system or expedient wastewater system, during the initial stages of an austere beddown operation, drainage pits and drainage trenches may be the only options to dispose of generated gray water. See **Chapter 4** for information on expedient gray water disposal measures.

2.7.1. Hand-washing stations. When manufactured field washstands are not available, create expedient or improvised hand-washing stations to enhance sanitation. Simple hand-washing devices should be available outside field latrine locations and dining facilities. These devices should be easy to operate and have a constant supply of water. Construct a soakage pit under each device to prevent water from pooling around the hand-washing station. As previously mentioned, emphasize the importance of using hand-washing devices to prevent disease transmission. Shown in **Figure 2.16** and **Figure 2.17** are examples of improvised and expedient hand-washing stations.

Figure 2.16. Improvised Hand-Washing Station.



Figure 2.17. Expedient Hand-Washing Station.



2.7.2. Expedient showers. For obvious sanitary reasons, bathing facilities should be available during field activities. Showers have many advantages over bathtubs;

specifically, they use significantly less water and provide better sanitation. When the field situation allows construction of expedient shower devices, locate shower stations in areas convenient for assigned personnel. Ensure showering devices have a soakage pit under them to prevent water from pooling. For privacy, add a curtain or screen around the shower station. **Figure 2.18** and **Figure 2.19** provide examples of expedient, rudimentary shower stations.

Figure 2.18. Single Station Expedient Shower.

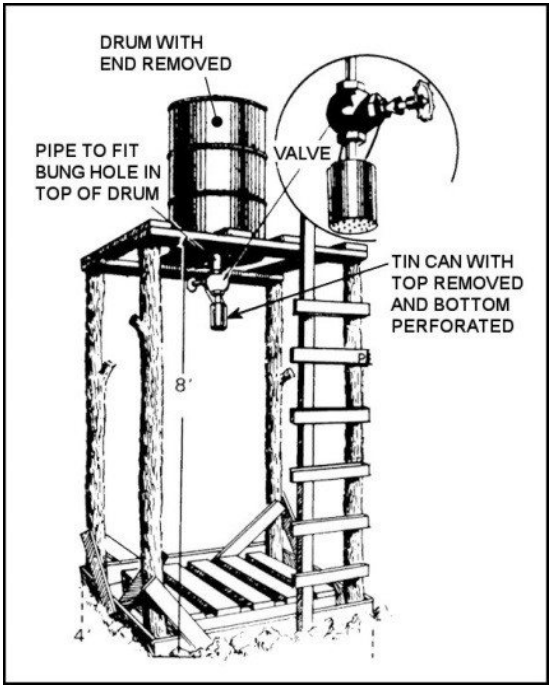
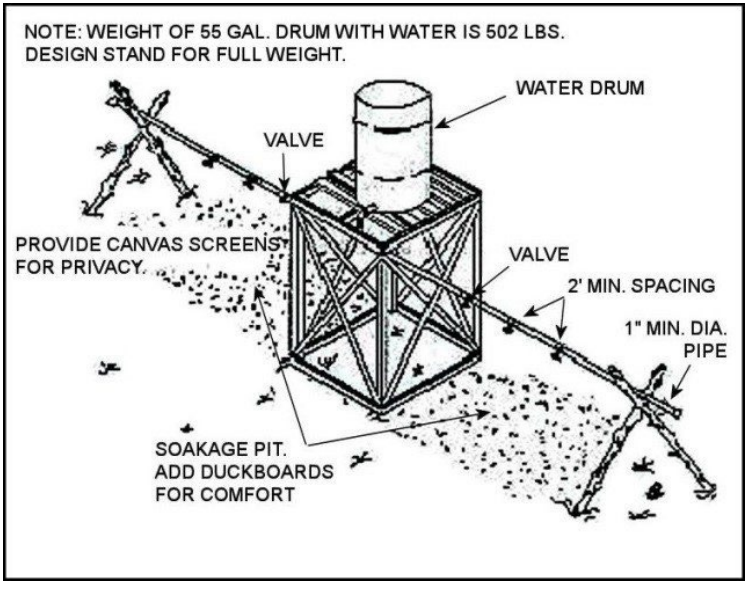
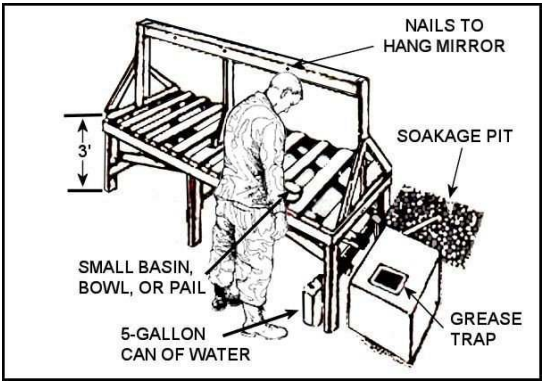


Figure 2.19. Multiple Station Expedient Shower.



2.8. Shave Stations. While it may be difficult maintaining Air Force appearance standards under field conditions, daily shaving and a beardless face is especially important when wearing a protective mask under chemical or biological warfare conditions. It is difficult to obtain an airtight mask seal with even stubble on the face. When field conditions permit, consider constructing expedient shave stations similar to that illustrated in **Figure 2.20** to support base residents. Discarded wastewater should pass through a grease trap before entering the soakage pit or trench to prevent clogging. See **Chapter 4** for more information on grease traps.

Figure 2.20. Expedient Shave Station.



2.9. Summary. This chapter presents methods for creating field-expedient sanitation and hygiene facilities at contingency locations. The facilities are temporary options until sustainable assets arrive. For additional information on field sanitation and waste disposal methods at contingency locations, consult the references listed in **Table 2.4** and other theater-specific guidance.

Table 2.4. Chapter 2 Quick References.

Field Expedient Hygiene and Sanitation	
DoDI 4715.19	*Use of Open-Air Burn Pits in Contingency Operations*
DoDI 4715.22	*Environmental Management Policy for Contingency Locations*
AFH 10-222V4	*Environmental Considerations for Overseas Contingency Operations*
AFTTP 3-32.34V1	*Civil Engineer Bare Base Development*
AFPAM 10-219V5	*Bare Base Conceptual Planning*

Chapter 3

EXPEDIENT WATER SOURCE DEVELOPMENT

3.1. General Information. The location of suitable water sources is usually identified before deploying and confirmed upon arrival at the site. Key factors include whether the water is fresh, brackish, or salt water; comes from a river, lake, ocean or well; its temperature; and distance from point of use. In some situations, available water sources need further cultivation or development before becoming a reliable source.

3.2. Overview. This chapter addresses expedient and improvised methods of developing surface water sources at austere, remote, or dispersed contingency locations when time and materials are limited, or when conventional methods are impractical to employ. Once water sources have been located for a beddown location, field engineers further develop the source for use as needed. This may include actions to increase water quantity, quality, or availability for treatment and distribution. Consult bioenvironmental engineering for water quality source assessment and monitoring. **Note:** Refer to AFMAN 48-138_IP, *Sanitary Control and Surveillance of Field Water Supplies*, AFH 10-222V4, AFPAM 10-219V5, and AFTTP 3-32.34V1 for more information on developing, treating, and managing water resources at contingency locations.

3.3. Surface Water Sources. Water from rivers, streams, lakes, or springs are the most common sources of surface water supply. Surface water is usually easy to find and may be easily developed. Ground water is accessed through springs or man-made wells. Consider developing ground water sources when surface water sources are inadequate or unusable. In regions such as tropical islands where there is abundant rainfall and rapid surface runoff, rainwater is the primary source for the inhabitants. However, the quantity of water available may not be sufficient to supply the needs of both the civilian population and military forces. Rainwater as a source may be sufficient for small units for limited operations, but it should not be considered if other more reliable sources are available.

3.3.1. Rivers, streams, and lakes. These water sources are usually capable of supplying adequate quantities of water that can be easily accessed with mobile water treatment systems. However, water quantity, quality, flow, depth, etc., could still be an issue at the intake point. Expedient and improvised dams, reservoirs, and intakes devices may help mitigate some of the issues in the short term.

3.3.1.1. Dams and reservoirs. If a small river or stream is the best available option, consider increasing its capacity to meet requirements by constructing a small expedient dam or reservoir. A typical layout of a small, expedient dam and reservoir is illustrated in **Figure 3.1** through **Figure 3.3**.

Figure 3.1. Expedient Dam and Reservoir Layout.

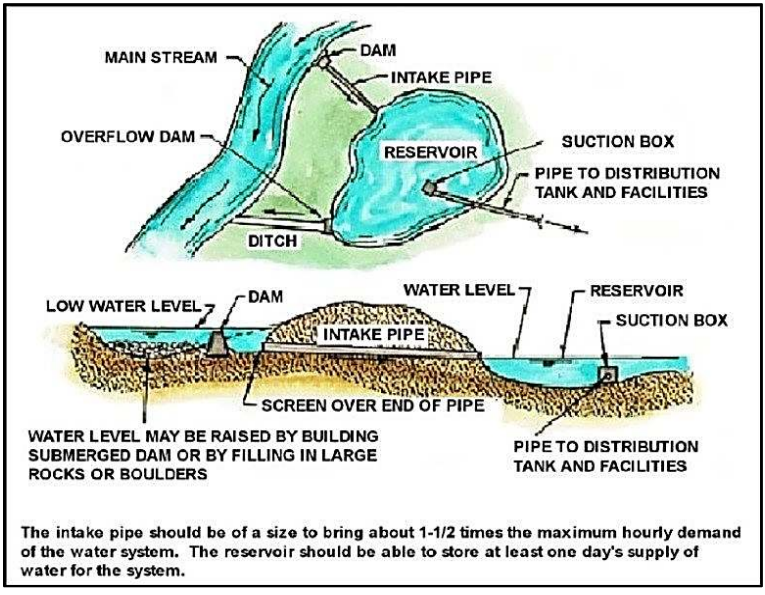


Figure 3.2. Expedient Dam for Impounding a Small Stream.

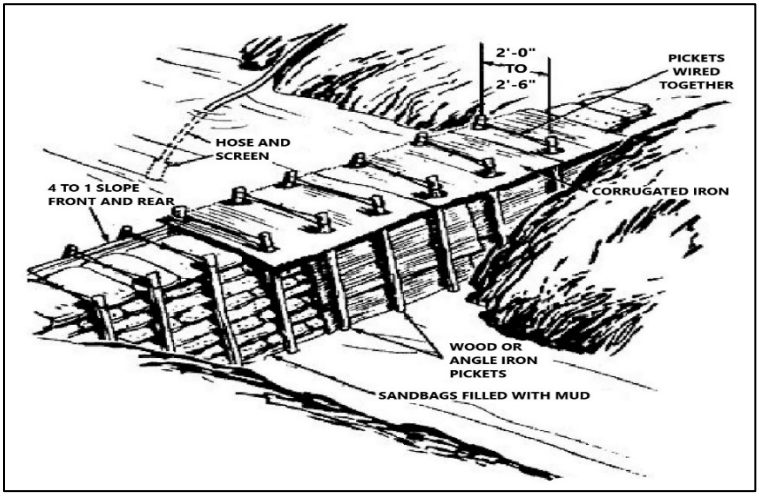
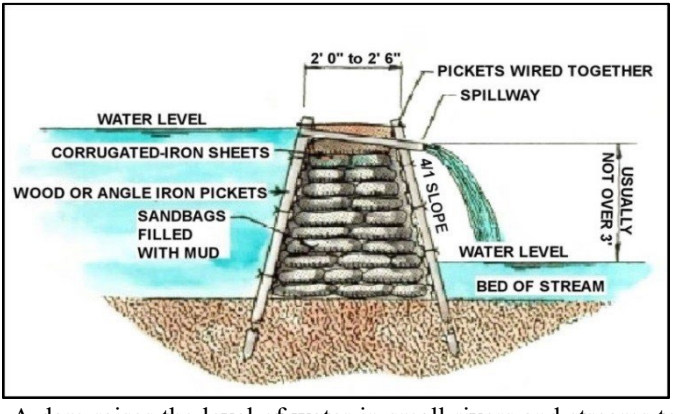


Figure 3.3. Expedient Small Dam (Side Profile).

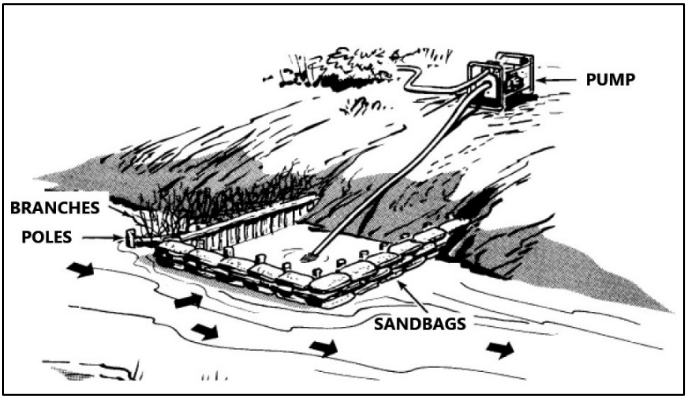


3.3.1.1.1. A dam raises the level of water in small rivers and streams to provide an adequate depth of cover for intake strainers. For swift flowing streams,

consider constructing a wing dam or baffle dam to slow water flow near the bank and protect the intake screen without completely stopping water flow. **Figure 3.4** is an example of a baffle dam.

3.3.1.1.2. Reservoirs store raw water for both firefighting and to use for purification and treatment into potable water. Once the natural water resources are confined in a reservoir, the water must be delivered to the point of usage. This is generally accomplished by pumps or gravity pipelines.

Figure 3.4. Expedient Baffle Dam.



3.3.1.2. Surface water intakes. Water at the intake point should be as clear and deep as possible. Install screens on intakes regardless of water appearance; they keep fish and debris out of the water system. Special attention must be given to the intake. If a lot of muck or silt are on the bottom of the water source, a floating type of intake is best. If the bottom of the water source is a bed of sand or gravel, a surface intake with a screen may be equally effective and easier to use.

3.3.1.2.1. Floatable intakes. Expedient floats made of logs (**Figure 3.5)**, lumber, sealed cans, or empty fuel drums (**Figure 3.6**) can support the intake strainer in deep water when other assets are not available. They are especially useful in large streams where the quality of water varies across its width or where water is not

deep enough near the banks to cover the intake strainer. An adequate depth of water can cover the intake point by anchoring or stationing the float at the deep part of the stream. Secure the intake hose to the top of the float, allowing enough slack for movement of the float. If support lines secure the float to the banks, alter the position of the float to correspond to changes in depth by manipulating the lines. The chief advantage of a float intake is easy adjustments of the screen. **Note:** Placing the strainer on the suction hose of a floating intake at least 12 inches below the water level reduces the likelihood of clogging the strainer with floating debris, or the loss of prime from air entering the suction line.

Figure 3.5. Log Float Water Intake.

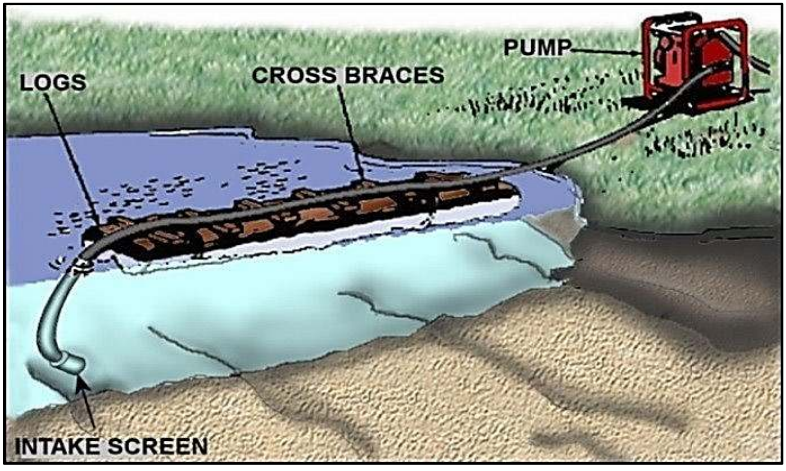
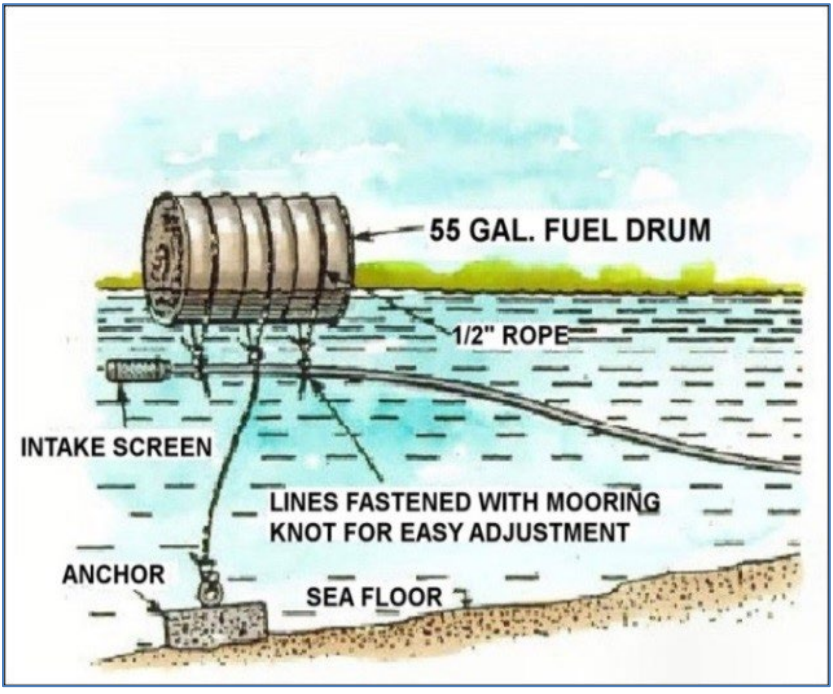
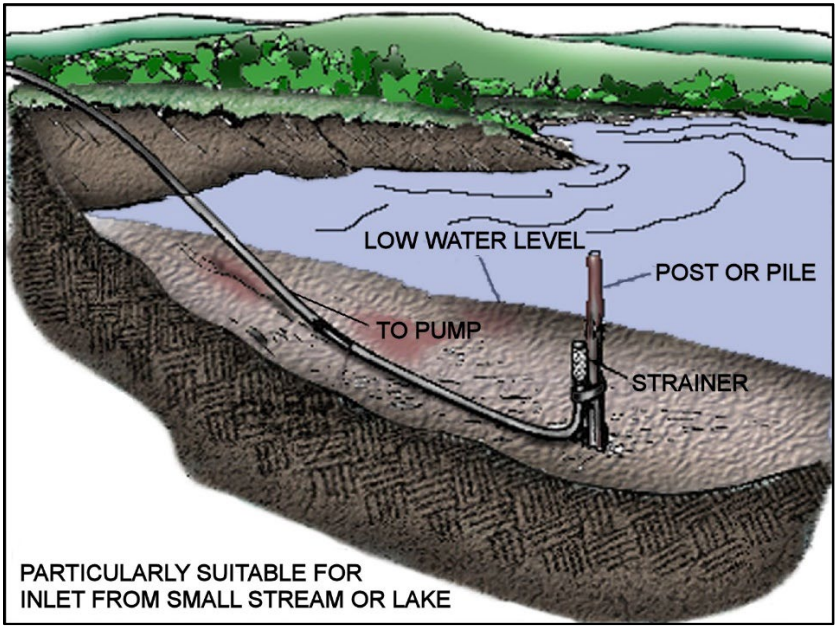


Figure 3.6. Drum Float Water Intake.



3.3.1.2.2. Rock and stake supported intakes. If the stream is not too swift and the water is deep enough, an expedient intake may be prepared by placing the intake strainer on a rock. This will prevent clogging of the strainer by the streambed and provide enough water above to prevent sucking air into the intake pipe. If the water source is a small stream or shallow lake, secure the intake pipe to a stake or post as shown in **Figure 3.7**.

Figure 3.7. Intake Hose Secured to Stake or Post.



3.3.1.2.3. Pit supported intakes. When a stream is so shallow that at least 12 inches of water does not cover the intake screen, dig a pit and lay the screen on a rock or board placed at the bottom of the pit. Line those pits dug in streams with clay or silt bottoms with gravel to prevent dirt from entering the purification equipment (**Figure 3.8**). Surround the screen with gravel to prevent collapse of the sides of the pit and shield the screen from damage by large floating objects. The gravel also acts as a coarse strainer for the water. Enclosing the intake screen in a bucket or other container as shown in **Figure 3.9** may also provide satisfactory results.

Figure 3.8. Intake Hose in Gravel Pit.

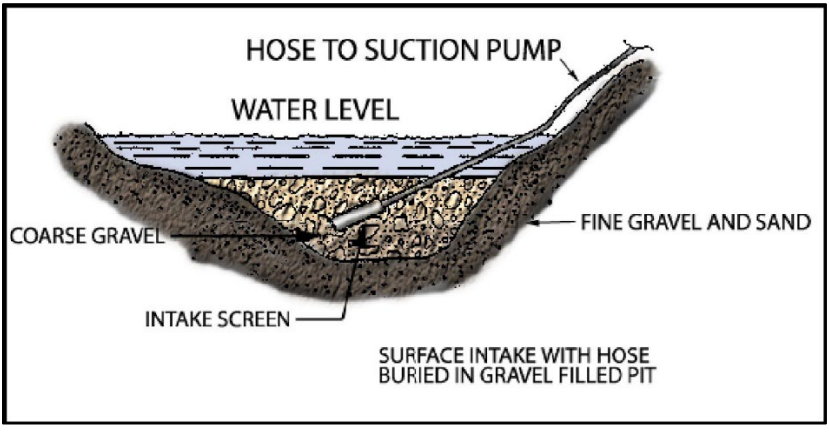
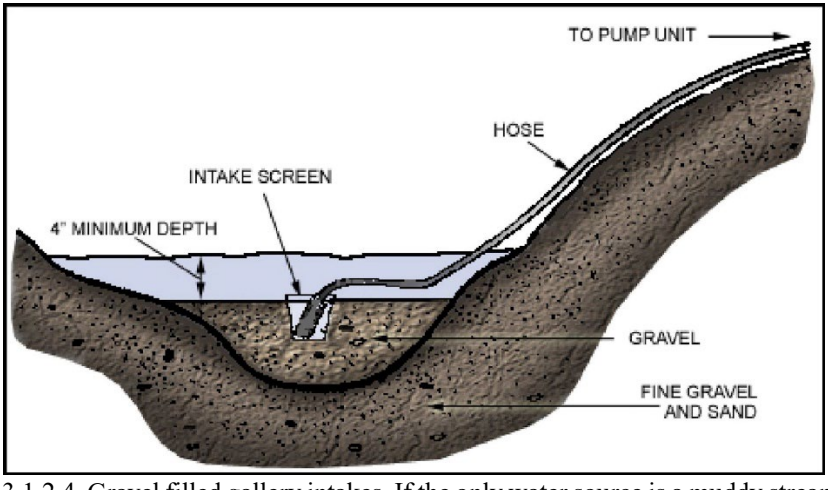


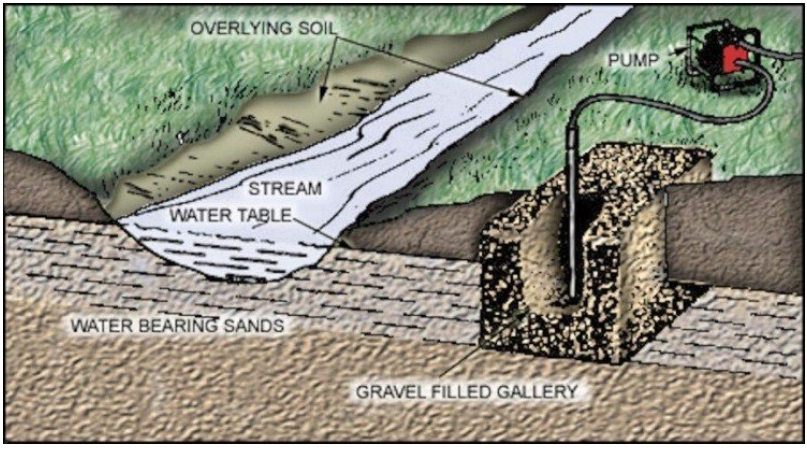
Figure 3.9. Intake Hose in Gravel Pit and Bucket.



3.3.1.2.4. Gravel filled gallery intakes. If the only water source is a muddy stream, the water quality can be improved by digging intake galleries along the bank. Dig

a trench along the bank deep enough so water from the stream percolates into it to intercept ground water flowing toward the stream. Fill the trench with gravel to prevent the sides from collapsing. Place the intake strainer in the gravel below the water line as shown in **Figure 3.10**. The amount of work required to produce the gallery is justified by a reduction in the chemicals needed to coagulate the water, the elimination of the necessity of frequently backwashing the filter, and the higher quality of water obtained.

Figure 3.10. Gravel Filled Gallery Intake.



3.3.2. Springs. Water emerging at the surface naturally with a distinct current is called a spring. When the current is not present, the flow is called a seep. Develop these sources to increase water quantity, quality, or availability for treatment and distribution. Some springs may be intermittent and depend on seasonal rain or snow. Consider developing springs and seeps that provide a dependable supply of water for anticipated need.

3.3.2.1. Spring development is generally accomplished by removing obstructions to the flow and providing a means to collect and store the water. It may be necessary to:

- Clear the spring of debris, undergrowth, topsoil, loose rocks, and sand to reduce pollution.
- Enlarge the outlet of the spring.
- Dig ditches or build pipeline tunnels to guide the water from spring to intake point.
- Construct a dam at the intake point to trap water from ditches or pipes.
- Build a collecting box (spring box).

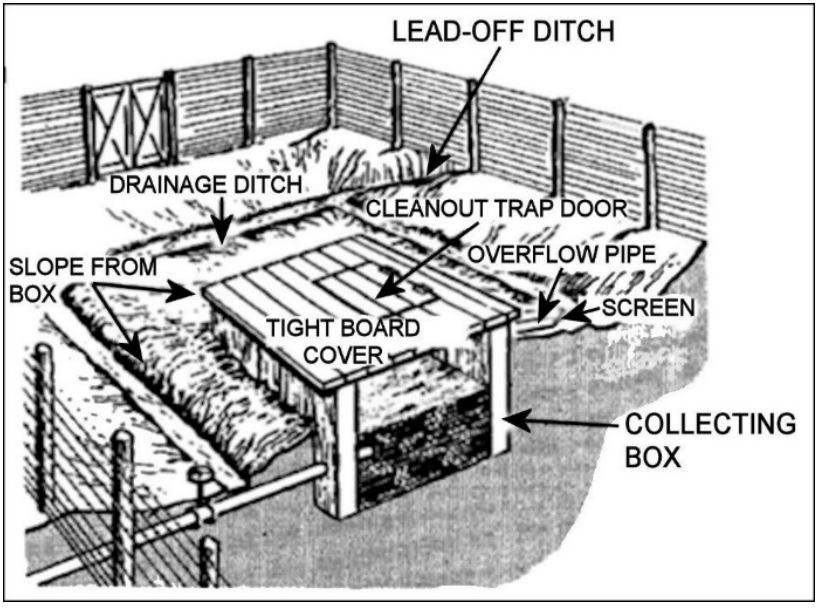
3.3.2.2. A collecting box (**Figure 3.11**) or basin can be made of wood, plastic, tile, or concrete. They collect water, which flows from rocks under the force of gravity. Collecting box design considerations include:

- The box should be large enough to hold most of the flow.
- Place it below the ground level so that only the top is slightly above the surface.
- Tightly cover the box to prevent contamination and decrease evaporation.
- Design the inlet to keep out surface drainage and prevent pollution.
- Fence the area and provide proper drainage.
- A screen on the overflow pipe keeps out insects and small animals.
- Another screen on the intake pipe keeps large, suspended particles from being taken in by the raw water pump.

3.3.2.3. Maintenance of the spring includes:

- Periodically removing accumulated sediment from the spring box.
- Keeping outlets and overflow pipes clear.
- Repairing rodent damage and erosion from overflow pipes.
- Keeping surface water diverted away from the spring.

Figure 3.11. Collecting Box.



3.4. Expedient Water Storage. During initial force beddown operations, items such as portable water tanks, onion bladders, fabric tanks, and improvised methods are options for expedient water storage. The examples shown in **Figure 3.12** and **Figure 3.13** are improvised, field-expedient water storage methods that can be implemented using local resources when traditional methods and assets are unavailable. Flexible bladders, water distribution trucks and trailers, and even sandbag or earthen berms lined with plastic sheeting are other possible alternatives worth considering. Regardless of storage methods used, ensure they are large enough to meet daily peak demands. To prevent potential sabotage by clandestine groups, storage facilities should be located well within the secure area of the base.

Figure 3.12. Expedient Water Storage Concepts.

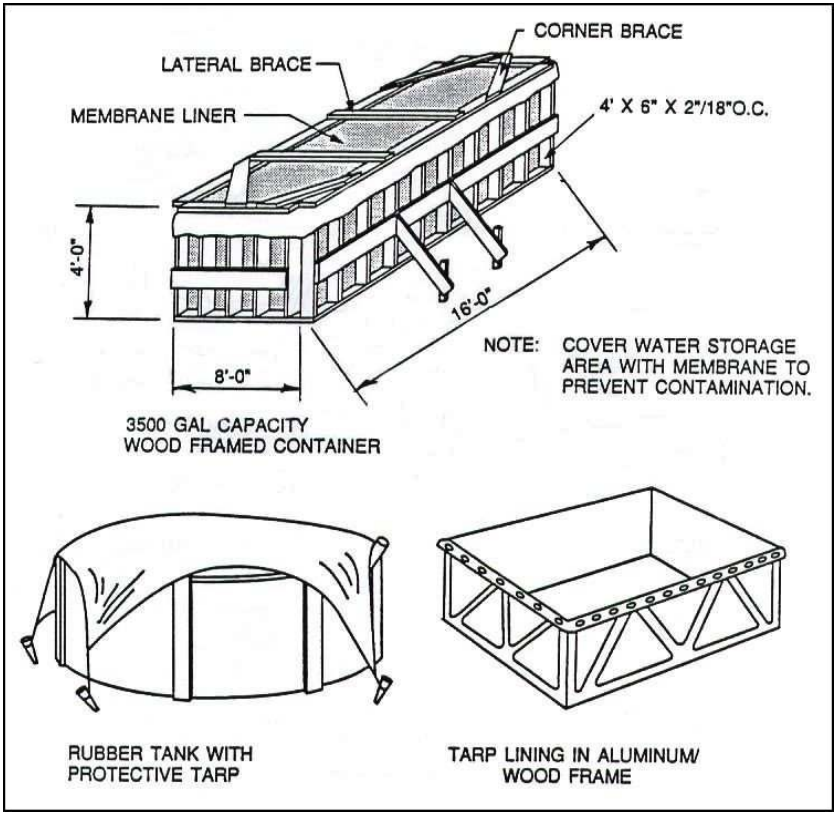
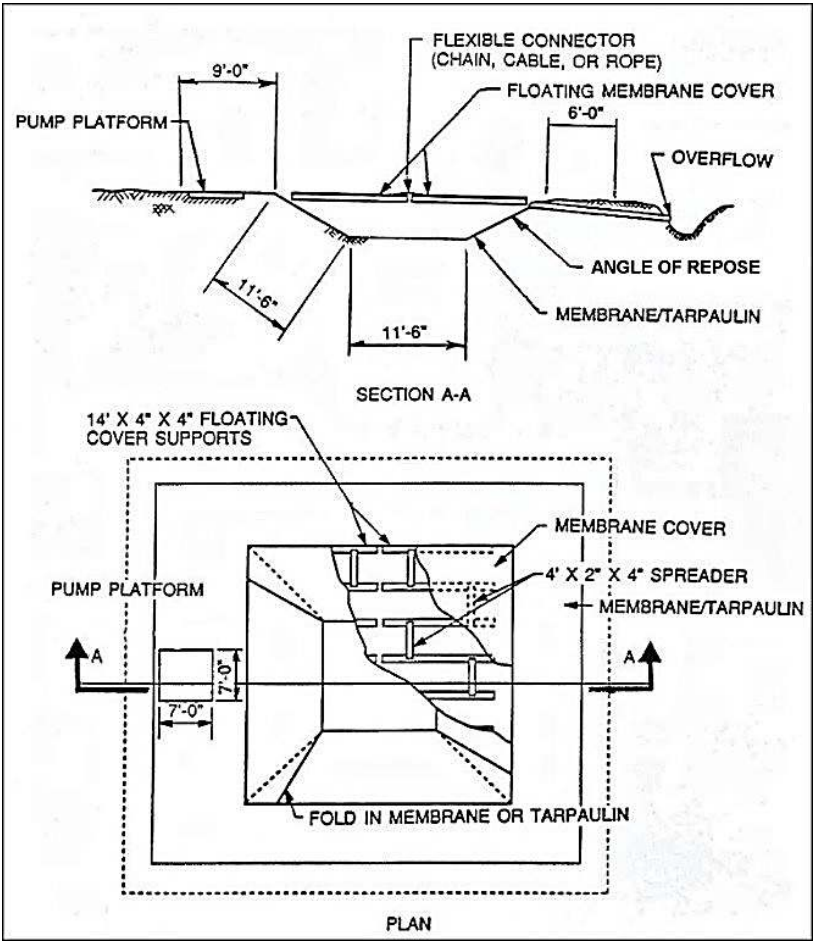


Figure 3.13. Earth Pit Water Storage.



3.5. Expedient Field Water Treatment. When cut off from supply lines and military-approved water is not available, select the clearest and cleanest water with the least odor available, and treat the water by boiling or using iodine tablets, Chlor-Floc dry calcium hypochlorite, or household bleach (sodium hypochlorite solution). Consult AFMAN 48-138_IP for details on the use of these and other field water treatment guidelines.

3.6. Summary. For beddowns at austere locations, drinking water is at the forefront of operational planning. Installation commanders take measures to ensure water sources are developed and potable water is readily available. It is a fundamental requirement for civil engineers to develop field water sources and produce potable water whenever needed. This chapter presented expedient options and ideas to help provide this capability. For additional information on development of water sources at contingency locations, consult the references listed in **Table 3.1** and other theater-specific guidance.

Table 3.1. Chapter 3 Quick References.

Development of Surface Water Sources	
DoDI 4715.22	*Environmental Management Policy for Contingency Locations*
AFH 10-222V4	*Environmental Considerations for Overseas Contingency Operations*
AFPAM 10-219V5	*Bare Base Conceptual Planning*
AFTTP 3-32.34V1	*Civil Engineer Bare Base Development*
AFMAN 48-138_IP	*Sanitary Control and Surveillance of Field Water Supplies*

Chapter 4

EXPEDIENT GRAY WATER DISPOSAL

4.1. General Information. Identifying and implementing wastewater disposal methods early can minimize unsanitary conditions that pose significant risks to personnel health and safety. For deployments to remote, austere, and dispersed contingency locations, consider using expedient methods to dispose of gray water until deployable systems become available or on-site systems are completed or restored. Consult public health or other aerospace medicine personnel when selecting sites for wastewater disposal activities. See AFH 10-222V4, AFPAM 10-219V9, and UFC 3-240-03, *O&M: Wastewater Treatment* for more information on wastewater disposal methods.

4.2. Overview. Gray water refers to wastewater from non-latrine sources such as showers, laundry, kitchens (except from food grinders/garbage disposals), vehicle wash racks, and hand-washing stations. This chapter addresses expedient disposal methods for gray water, including employment of kitchen soakage pits and trenches, evaporation beds, collection ponds, and grease traps.

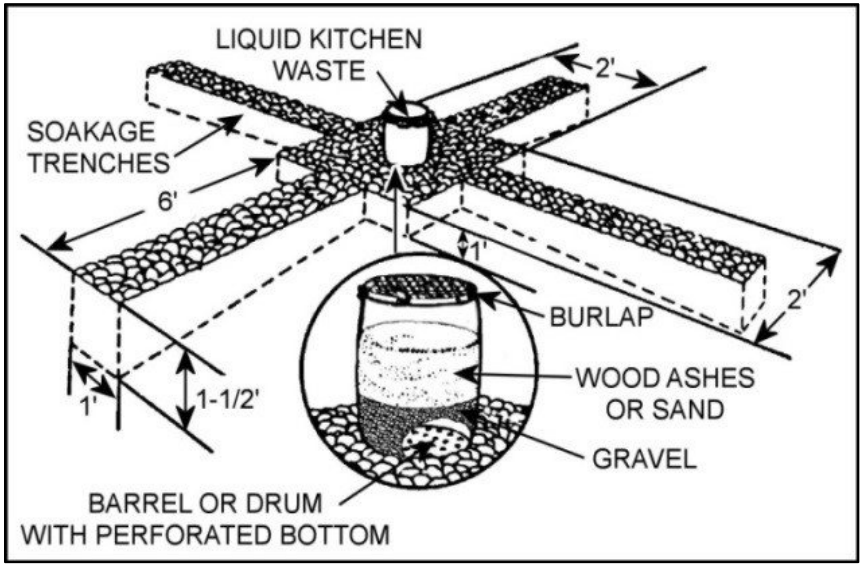
4.3. Kitchen Soakage Pits and Trenches. Beddown locations may require soakage pits or soakage trenches to dispose of wastewater from kitchens and dining facilities (DFAC) for short periods until a more suitable means of disposal is available. Climate and soil conditions usually determine the effectiveness of these devices. Normally, pits and trenches dispose of small volumes of wastewater, but with proper design and operation, they can be effective for larger volumes. Pretreatment using grease traps is necessary to remove grease, particulate, and organic material to prevent prematurely clogging the soakage pit.

4.3.1. Soakage Pit. Kitchen soakage pits constructed the same way as the urine soakage pit addressed in **paragraph 2.6.1**, will normally dispose of liquid kitchen waste for 200 persons. A grease trap replaces pipes or troughs used in the urine soakage pit. If deployment is longer term (e.g., last for several weeks), make two kitchen soakage pits and alternate their usage on a daily basis; giving soakage pits a rest period helps prevent clogging. Properly close any clogged soakage pits.

Note: To close a kitchen pit, backfill and compact with soil 1 foot above the grade and mark the pit with a rectangular sign indicating the type of pit and date of closure.

4.3.2. Soakage Trench. Use a kitchen soakage trench when the groundwater level or a rock formation precludes digging a soakage pit. The soakage trench has a 2 feet square and l foot deep pit (**Figure 4.1**). The pit has a trench radiating outward from each side for a distance of 6 or more feet. Dig trenches 1-foot wide, varying depth from 1 foot at the center to 1-1/2 feet at the outer ends. Fill the pit and trenches with material similar to that used in the soakage pit. Build two units for every 200 persons fed and alternate their usage daily. Use a grease trap with the soakage trench and close it the same as the soakage pit in **paragraph 4.3.1**.

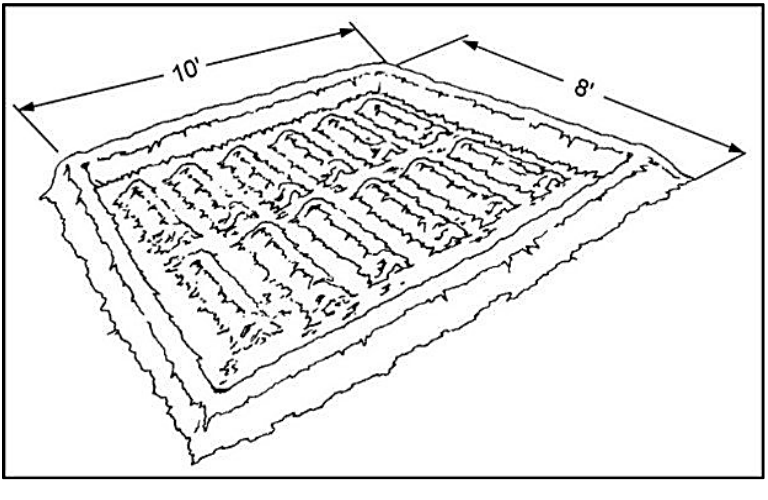
Figure 4.1. Soakage Trench.



4.4. Gray Water Evaporations Beds. The evaporation bed is a field expedient method to dispose of small flows of gray water. This method is suited for hot and dry climates and locations where clay soil prevents use of standard soakage pits. Although simple to build, they can be labor intensive to maintain. Beds should allow 3 square feet per person per day for kitchen waste and 2 square feet per person per day for wash and bath waste.

4.4.1. Normally, expedient evaporation beds measuring 8 by 10 feet (**Figure 4.2**) should be sufficient for small flows, but for larger flows, size the beds based on anticipated flow. As an example, a BEAR kitchen configured to serve 1,100 personnel would require 3,300 square feet of evaporation bed. If divided into seven individual beds of 472 square feet, each bed would measure 22 feet by 22 feet. Each bed should be about 0.62 feet deep, but to allow for free board each bed should be one foot deep.

Figure 4.2. Evaporation Bed.



4.4.2. Construct beds by scraping the topsoil to the edges, forming a small dike around the bed. Spade earth in the bed to a depth of 10 to 15 inches and rake soil

into a series of rows with the ridges approximately 6 inches above the depression. Form rows either lengthwise or crosswise, depending on which allows for best water distribution. If multiple beds, space beds to allow waste distribution to any one bed.

4.4.3. During the day, flood one bed with liquid waste to the top of the ridges. This is equivalent to an average depth of 3 inches over the bed. Allow liquid waste to evaporate and percolate. After three or four days, the bed is usually sufficiently dry for re-spading and reforming. Flood other beds on successive days and follow the same sequence of events.

4.4.4. Ensure proper bed rotation, maintenance, and dosage. It is essential that kitchen wastewater runs through an efficient grease trap (see **paragraph 4.7**) before flowing to an evaporation bed. If used properly, evaporation beds create no insect hazard and only a slight odor. Consider other potential waste disposal methods if they are more adaptable to the particular situation.

4.5. Gray Water Collection Ponds. If planning to use gray water for activities that do not require potable water such as dust suppression, irrigation, construction, or other allowed uses, consider collecting gray water for future use in a collection pond (**Figure 4.3**) or field-expedient wastewater collection box (see **paragraph 4.6**). Depending on usage, gray water may need to be treated before using it for other purposes. For more details on field expedient collection and disposal of gray water, see AFTTP 3-32.34V1, AFPAM 10-219V5, and AFPAM 10-219V9. **Note:** It is important to remember the hazards of storing gray water for extended periods. If allowed to sit, gray water can turn into stagnant, sludge-filled concoctions of bacteria and pathogens—it can essentially become black water in as little as 24 hours if not filtered and treated or disposed of properly.

Figure 4.3. Gray Water Collection Pond.



4.6. Field-Expedient Wastewater Collection Pit Box. During initial stages of beddown operations, certain facilities may generate wastewater requiring temporary collection until disposed of properly. While this situation is uncommon, a pit box offers an expedient, temporary means to collect wastewater should the need arise. A 1,000-gallon pit box should be adequate for facilities producing relatively small quantities of wastewater. Facilities producing larger amounts of wastewater may require a 2,000-gallon pit box. **Table 4.1** and **Table 4.2** list building materials and **Figure 4.4** provides construction details for pit boxes. Empty collection pits using wastewater disposal trailers and trucks.

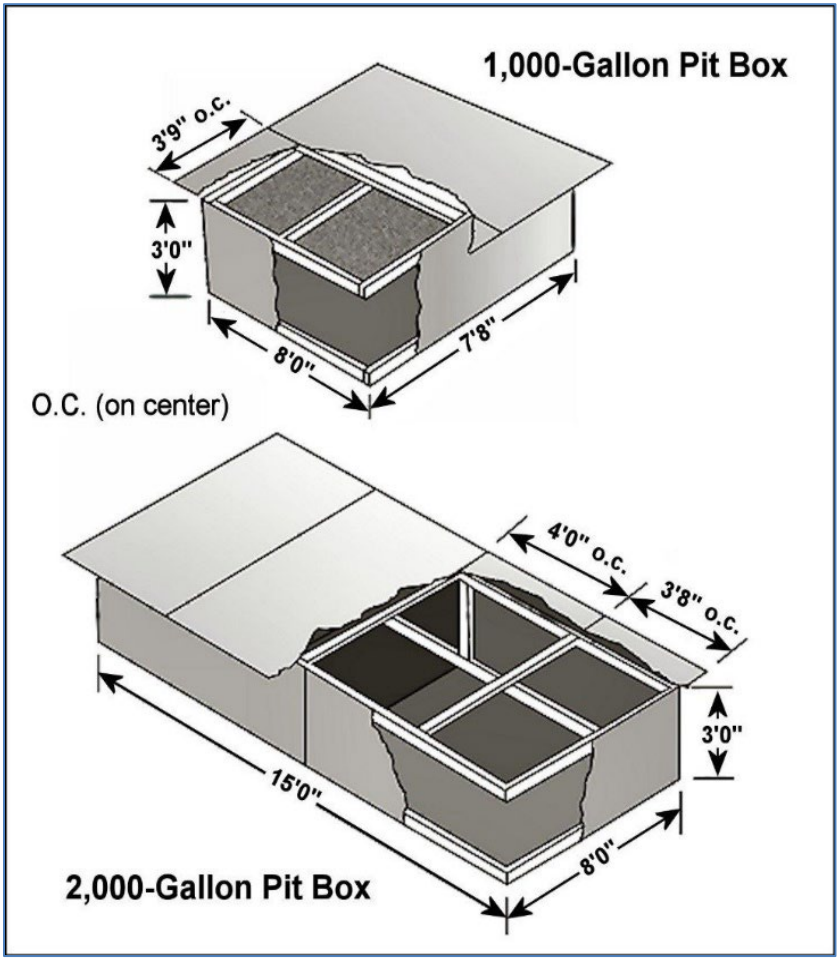
Table 4.1. Materials for 1,000-Gallon Wastewater Collection Pit Box.

1,000-Gallon Pit Box		
Description	Number Required	Remarks
1/2″ x 6′-0″ x 7′-6″	1 ea.	Floor
1/2″ x 3′-0″ x 7′-6″	2 ea.	Side Panel
1/2″ x 3′-0″ x 6′-0″	2 ea.	End Panel
2 x 4 x 7′-6″	4 ea.	Framework
2 x 4 x 5′-8 3/4″	4 ea.	Framework
2 x 4 x 5′-8 3/4″	1 ea.	Brace*
15′ x 15′ Sheet Plastic	1 ea.	Liner
Nails, sizes 6d / 10d	1.00 lb. each size	
***Line box with plastic before installing braces.**		

Table 4.2. Materials for 2,000-Gallon Wastewater Collection Pit Box.

2,000-Gallon Pit Box		
Description	Number Required	Remarks
1/2″ x 6′-0″ x 7′-6″	2 ea.	Floor
1/2″ x 3′-0″ x 7′-6″	4 ea.	Side Panel
1/2″ x 3′-0″ x 6′-0″	2 ea.	End Panel
2 x 4 x 15′-0″	4 ea.	Framework
2 x 4 x 5′-8 3/4″	4 ea.	Framework
2 x 4 x 5′-8 3/4″	2 ea.	Upright
2 x 4 x 5′-8 3/4″	3 ea.	Brace*
1/2″ x 4′-0″ x 8′-0″	4 ea.	Cover
15′ x 15′ Sheet Plastic	1 ea.	Liner
Nails, sizes 6d / 10d	1.50 lb./ 2.00 lb.	
***Line box with plastic before installing braces.**		

Figure 4.4. Construction of Wastewater Collection Pit Boxes.



4.7. Expedient Grease Traps. Whenever grease-laden wastewater is disposed of into an expedient wastewater septic system, grease traps should be installed before the septic tank or separate drain field to prevent clogging system components. Grease traps come in countless shapes and sizes, but their operation is basically the same. Just as the name suggests, a grease trap catches grease (animal fats and oils) from wastewater. Although there are many different models, grease traps are typically placed in the flow line of the building's sewer system to catch grease and fats preventing clogs in the waste pipes. The following paragraphs address construction details for various field-expedient grease traps used with contingency wastewater treatment and disposal methods.

4.7.1. In-line grease traps (or interceptors). The most effective in-line grease traps are usually located outside, in an underground tank with ground-level access. These devices should be watertight with inlet and outlet piping. The materials used for grease trap construction vary, including concrete, steel, fiberglass, plastic polymer, or other material. As illustrated in **Figure 4.5** and **Figure 4.6**, wood and 55-gallon drums are optional field expedient or improvised methods for constructing in-line grease traps.

Figure 4.5. Wooden-Sheathed Grease Trap.

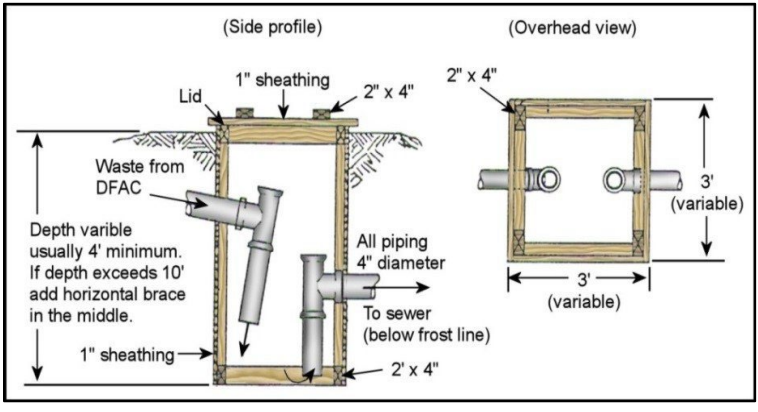
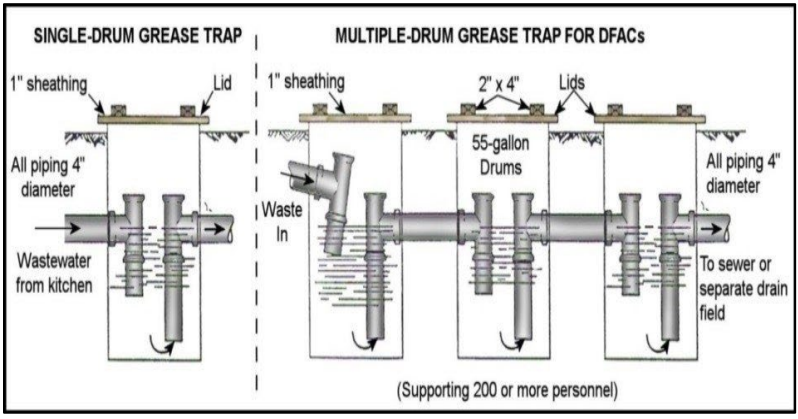


Figure 4.6. 55-Gallon Drum Grease Trap.



4.7.2. Barrel-filter grease trap. Construct the expedient barrel-filter grease trap from a 30- to 55-gallon barrel or drum as addressed below and illustrated in **Figure 4.7**:

- Remove the barrel top and bore a number of large holes into the bottom.
- Place 8 inches of gravel or small stones in the bottom of the barrel and cover them with 12 to 18 inches of wood ashes or sand.
- Next, fasten a piece of burlap to the top of the barrel to serve as a coarse strainer.
- Place the trap directly over the soakage pit or on a platform with a trough leading to the pit. Be sure to empty the trap every two days.
- Wash the trap, remove and bury the ashes or sand, and refill the trap with fresh ashes or sand.
- Also, wash the burlap strainer every day or replace it. **Figure 4.8** depicts two additional versions of expedient barrel-type grease traps.

Figure 4.7. Barrel-Filter Grease Trap.

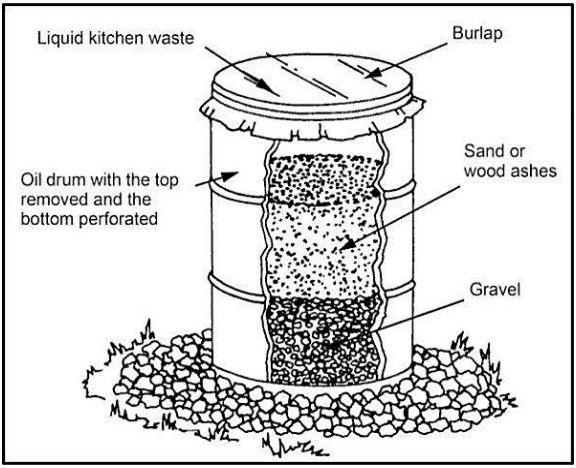
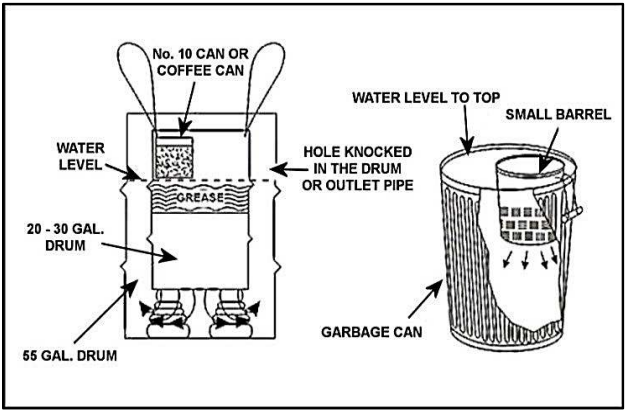


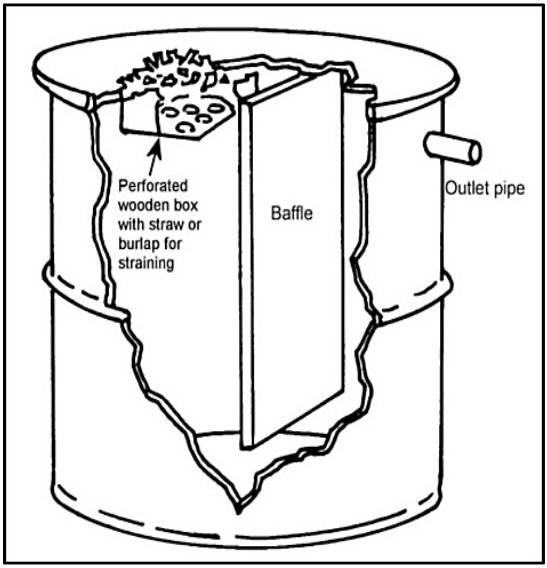
Figure 4.8. Expedient Barrel and Garbage Can Grease Traps.



4.7.3. Baffle grease trap. Construct a baffle grease trap from a 55-gallon drum or box (**Figure 4.9**). Divide the trap vertically into unequal chambers with a wooden

baffle. This baffle should extend to within 1 inch from the bottom. Waste pours through a strainer into the large chamber, then passes under the baffle and flows into the small chamber. The large chamber should have a removable lid and strainer. The strainer may be a box with openings in the bottom. Fill the strainer with straw or burlap to remove coarser solids. Clean the strainer frequently by scrubbing it with soap and water to prevent clogging. Insert a 1-inch pipe 3 to 6 inches below the top of the smaller chamber to carry liquid from the trap to the soakage pit. Clean the trap frequently to ensure proper operation. Remove the grease, drain the trap, and remove the sediment from the bottom. Burn or bury the grease, sediment, and strained material.

Figure 4.9. Baffle Grease Trap.



4.7.4. Dining facilities (DFAC) create a significant amount of grease-laden wastewater, and the grease and food particles present a complicating factor for wastewater treatment and disposal. It is vital to use and properly maintain grease traps when discarding wastewater from DFACs into soakage pits and trenches.

4.8. Summary. This chapter presented expedient options for sewage and wastewater disposal during contingencies. While not applicable in every situation, some methods and techniques may be useful as interim measures until prepackaged contingency materials, equipment and supplies arrive; existing systems repaired; or engineers build a more sustainable, wastewater treatment system. For additional information on sewage and wastewater disposal techniques and procedures at contingency locations, consult the references listed in **Table 4.3** and other theater specific guidance.

Table 4.3. Chapter 4 Quick References.

Sewage and Wastewater Disposal	
AFH 10-222V4	*Environmental Considerations for Overseas Contingency Operations*
AFPAM 10-219V5	*Bare Base Conceptual Planning*
AFPAM 10-219V9	*Semi-Permanent Wastewater Treatment Plants at Contingency Locations*
AFTTP 3-32.34V1	*Civil Engineer Bare Base Development*
UFC 3-240-03	*O&M: Wastewater Treatment*

Chapter 5

EXPEDIENT AND SHORT-TERM SOLID WASTE DISPOSAL

5.1. General Information. Properly disposing of solid waste (SW) can be challenging at austere beddown locations. A large volume of SW is generated during force beddown. Materials such as wood pallets and crates; scrap building materials; cardboard boxes; food waste and packaging; and many other paper, wood, plastic, and metal remnants are part of the waste stream. This waste can create excellent breeding grounds for rodents and insects if not properly managed and disposed of expeditiously. Not only does overall health and environmental protection necessitate proper waste disposal, the safety risks to personnel from inadequate or improper waste disposal is very high. For contingency locations, DoDI 4715.22, DoDI 4715.19, and CCMD directives and guidance apply. Additionally, the Solid Waste Management Plan (SWMP) for each contingency operation provides detailed SW management procedures.

5.2. Overview. Local SW disposal options usually depend on the capabilities and services available at each contingency location. This chapter addresses expedient methods to dispose of small quantities of non-hazardous SW for short periods, and the restricted use of open-air burn pits. When consistent with the SWMP, these methods can be useful interim measures until alternative disposal options become available, including reusing, repurposing, and recycling. Review AFTTP 3-32.34V1 for more information on SW disposal at contingency locations,

5.3. Expedient Solid Waste Disposal. During the initial days of beddown at bases in austere locations, it is unlikely a fully organized SW collection and disposal operation will be in place. Even if contract SW services are used, it could take several weeks to achieve full service. Furthermore, interruption of such services is likely during periods of increased security, particularly after an attack against the base. At all contingency locations (initial, temporary, and semi-permanent), backup expedient SW disposal plans should be available in case of interruption to regular or conventional methods. Airmen should perform basic SW disposal duties to maintain health of the force. Ensure improvised SW disposal methods comply with CCMD guidance.

5.3.1. Expedient SW disposal methods include garbage burial pits and trenches, barrel and mobile incinerators, and burn pits. **Note:** DoW policy imposes restrictions on the use of burn pits (see DoDI 4715.19). For those locations with more than 100 attached or assigned personnel in place for more than 90 consecutive days, adherence to DoDI 4715.19 Appendix 3A, Open-Air Burn Pit Health Risk Assessment, applies and must be coordinated with the bioenvironmental engineer.

5.3.2. As a general rule, SW disposal activities should be sited in the predominate downwind direction and a safe horizontal distance from personnel and water sources, away from areas with high water tables, channels, creeks, and areas significantly impacted by storm water runoff (always consult public health or other aerospace medicine personnel when siting these activities). Consider soil characteristics; look at topography, soil permeability, potential for runoff, etc.

5.3.3. Keep in mind SW activities attract rodents, snakes, etc., and must be considered in pest management plans and coordinated with the public health staff.

5.4. Garbage Burial Pits and Trenches. Garbage burial pits and trenches are interim measures (normally 10 days or less) to handle small quantities of SW during initial base beddown or other contingency operations until sustained SW disposal capabilities are in place (i.e., landfills, incinerators, contracted, etc.). Review SWMP for specific waste disposal guidance and requirements.

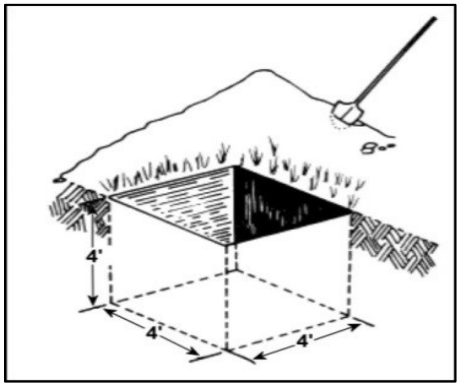
5.4.1. For small groups and short-term operations, consider burying garbage in pits approximately 4 feet square and 4 feet deep (**Figure 5.1**). Fill these pits no closer than 1 foot below ground surface, spray with insecticide, and mound over with compacted earth to 1 foot above ground level. Mark each pit with a rectangular sign on top of the mound indicating the type of pit and date closed. When larger pits are used, cover the garbage daily. Insecticide is not required if 2 feet of compacted soil are used for cover.

5.4.2. For larger groups or extended periods/operations, consider burying garbage in a continuous trench or combine with other solid waste in an established sanitary

landfill. A continuous trench can accommodate a large amount of garbage and allows for a daily coverage of earth as the trench is extended. The trench is first dug about 2 feet wide, 3 to 4 feet deep, and long enough to accommodate the garbage for the first day. As in the pit method, fill the trench to no closer than 1 foot from the top. Extend the trench as required and use the excavated dirt to cover and mound the garbage already deposited. Repeat this procedure daily or as often as garbage is dumped.

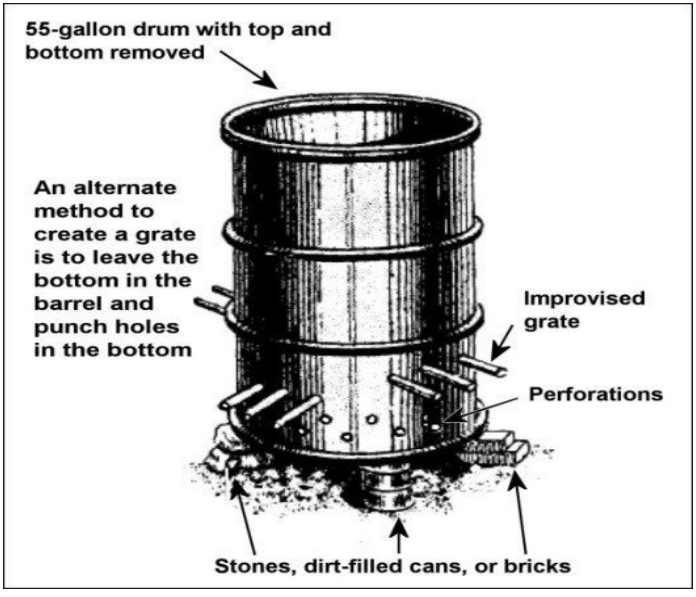
5.4.3. When using burial pits and trenches, closely monitor "self-help" garbage burial operations to prevent uncontrolled refuse disposal, which can lead to serious fire and sanitation hazards. Also, establish a stringent pest control program to alleviate danger of disease spread.

Figure 5.1. Garbage Burial Pit.



5.5. Expedient Barrel Incinerators. Expedient barrel incinerators should only be used in emergency situations until approved incinerators or other SW disposal methods are implemented. Barrel incinerators may be useful for small group, remote or dispersed location operations if the tactical situation allows. Typically constructed from 55-gallon drums, the barrel incinerator (**Figure 5.2**) can handle relatively small amounts of garbage. Follow CCMD guidance and SWMP for all burning operations.

Figure 5.2. Expedient Barrel Incinerator Details.



5.5.1. Before modifying the drum, remove toxic, flammable, or combustible contents. Like other improvised SW disposal methods, use barrel incinerators in temporary/emergency situations only if allowed by SWMP.

5.5.2. Conduct burn operations where prevailing winds carry smoke away from troop locations and living areas. Do not incinerate hazardous waste; explosions or toxic gases may result in injury to personnel or equipment damage.

5.6. Mobile Incinerators. The air curtain burner or incinerator (burn box) shown in **Figure 5.3** is one type of mobile incinerator used for expedient SW disposal at contingency locations. Other incinerators include those designed especially for medical, explosive, and ammunition waste. Air curtain incinerators burn more efficiently and produce lower smoke emissions than open-air burn pits.

5.6.1 The incinerator introduces a controlled velocity of air across the upper portion of a combustion chamber in which the waste is loaded. The curtain of air created in this process traps unburned particles under the curtain in the high temperature zone where temperatures can reach 2000º F. Increased combustion time and turbulence of air curtain burning results in a re-burn and more complete combustion of the loaded waste. Because incinerators operate at extremely high temperatures, exercise caution to avoid injury. Make sure personnel have the appropriate PPE.

5.6.2. As with other SW disposal methods, consider how waste for incinerators will be collected, transported, stored, etc. Ensure non-combustibles and waste prohibited from incineration by the SWMP are collected separately; do not deliver this waste to the incinerators. Also, ensure SW is sorted to remove ammunition, glass, batteries, metals, and most wood products.

Figure 5.3. Air Curtain Incinerators at Forward Location.



5.7. Open-Air Burn Pits. DoW policy imposes restrictions on the use of open-air burn pits. When consistent with policy in DoDI 4715.19, DoDI 4715.22, and CCMD guidance, use open-air burn pits as a short-term solution during contingency operations where no alternative is feasible. For the longer term,

incinerators, engineered landfills, or other accepted SW management practices must be used whenever feasible. Only operate open-air burn pits in a manner that prevents or minimizes risks to the human health and safety of DoW personnel.

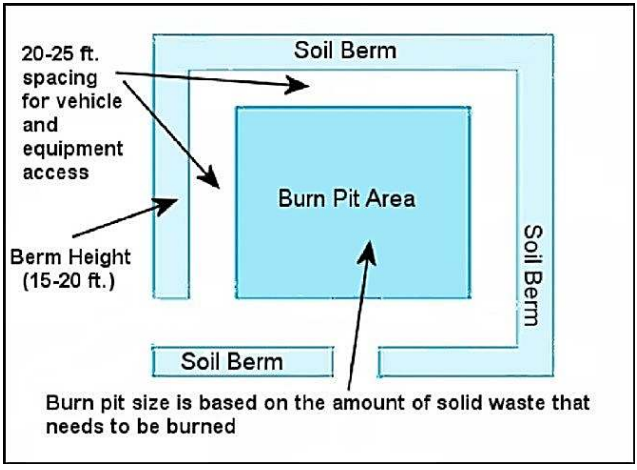
5.7.1. When open-air burn pits are used, SW plans should include provisions to transition to a more acceptable means of disposal. This becomes particularly important as bases mature and populations surge. Although burn pits reduce the overall waste volume and may be an effective means of disposing of certain waste streams, they can present health and quality-of-life issues. They must be operated in accordance with the CCMD and SWMP guidance and in a manner that prevents or minimizes risks to human health and safety of base personnel and where possible, harm to the environment.

5.7.2. Open-air burn pits should be located so that prevailing winds carry smoke away from the base or encampment, including housing, medical facilities, DFACs, and other troop concentrations. Remember to monitor the content entering the burn pit to prevent the disposal of hazardous waste/hazardous materials and other restricted waste products. Keep live ammunition and batteries out of SW burn pits.

5.7.3. When constructing the burn pit, determine the size by the amount of solid waste that needs burning. Place a soil or earthen berm around the burn pit for safety. **Figure 5.4** illustrates an example of a burn pit that uses a soil berm.

5.7.3.1. Determine the height of the berm (usually 15’-20’ high) based on factors at the beddown location. Factors such as groundwater table, pit design (above or below ground, loading ramps), maintenance equipment, personnel safety, etc., may affect the required height of the berm.

Figure 5.4. Layout of Open-Air Burn Pit with Soil Berm (Notional).



5.7.3.2. Although berms for burn pits are similar to other expedient berms, other design and construction features may be incorporated. For example, added dirt ramps allow access by dozers and compactors to perform routine maintenance on the pit, and additional fencing around berms to contain blowing trash and prevent unwanted entry. See AFTTP 3-32.33V2, *Expedient Roads, Drainage, and Protective Structures*, for specific berm construction methods.

5.8. Landfills. Constructing new sanitary landfills at contingency locations is not considered an expedient or short-term SW disposal method but using an existing landfill within reasonable distance and with a capacity to accept additional solid waste could be an available expedient option. Most landfills can accommodate a wide variety of SW types; except for hazardous waste and materials, nearly all rubbish, garbage, trash, ashes, solid organic waste, and miscellaneous solids may be disposed of safely. Consider using preexisting landfills whenever possible; contracting representatives can provide more information on contracted SW services. If establishment of an engineered landfill at your location is being considered, check theater SWMP for guidance and UFC 3-240-11, *Landfills In Support of Military Operations*, for landfill design/construction specifications.

5.9. Summary. During initial bare-base development, expedient SW disposal methods (e.g., burial pits, burial trenches, burn barrels, burn pits) may be required until other methods are established. If contract disposal is unavailable, engineers implement methods to manage SW disposal consistent with theater SWMP. If external disposal sites are out of service or inaccessible, identify internal locations. Garbage and debris can either be buried, burned, or a combination of both. It is important to consider all available methods of solid waste disposal, including incinerators, landfills, open-air burn pits, composting, and recycling facilities. When evaluating options, recognize that the operation of open-air burn pits involves severe restrictions. For additional information on SW disposal tactics, techniques, and procedures, consult references listed in **Table 5.1**.

Table 5.1. Chapter 5 Quick References.

Solid Waste Disposal	
DoDI 4715.19	*Use of Open-Air Burn Pits in Contingency Operations*
DoDI 4715.22	*Environmental Management Policy for Contingency Locations*
UFC 3-240-11	*Landfills In Support of Military Operations*
AFH 10-222V4	*Environmental Considerations for Overseas Contingency Operations*
AFTTP 3-32.33V2	*Expedient Roads, Drainage, and Protective Structures*
AFTTP 3-32.34V1	*Civil Engineer Bare Base Development*

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Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DoDI 4715.19, *Use of Open-Air Burn Pits in Contingency Operations*, 11 March 2018

DoDI 4715.22, *Environmental Management Policy for Contingency Locations*, 18 February 2016

AFI 10-209, *RED HORSE Program*, 7 January 2026

AFI 10-210, *Prime Base Engineer Emergency Force (BEEF) Program*, 25 October 2023

AFI 33-322, *Records Management and Information Governance Program*, 23 March 2020

DAFMAN 91-203, *Department of the Air Force Occupational Safety, Fire, and Health Standards, 24 February 2026*

AFMAN 48-138_IP, *Sanitary Control and Surveillance of Field Water Supplies*, 1 May 2010

AFH 10-222V4, *Environmental Considerations for Overseas Contingency Operations*, 1 September 2012

AFPAM 10-219V5, *Bare Base Conceptual Planning*, 30 March 2012

AFPAM 10-219V9, *Semi-Permanent Wastewater Treatment Plants at Contingency Locations*, 7 December 2023

AFTTP 3-32.33V2, *Expedient Roads, Drainage, and Protective Structures*, 2 August 2018

AFTTP 3-32.34V1, *Civil Engineer Bare Base Development*, 20 October 2023

UFC 3-240-03, *O&M: Wastewater Treatment*, 3 October 2019

UFC 3-240-11, *Landfills in Support of Military Operations*, 24 March 2021

Prescribed Forms

None

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication.*

Abbreviations and Acronyms

AF—Air Force

AFCEC—Air Force Civil Engineer Center

AFH—Air Force Handbook

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFPAM—Air Force Pamphlet

AFTTP—Air Force Tactics, Techniques and Procedures

BEAR—Basic Expeditionary Airfield Resources

CE—Civil Engineer

CCMD—Combatant Command

DoD—Department of Defense

DoDI—Department of Defense Instruction

DoW—Department of War

IP—Interservice Publication

O&M—Operations and Maintenance

OPR—Office of Primary Responsibility

PPE—Personal Protective Equipment

SW—Solid Waste

SWMP—Solid Waste Management Plan

UFC—Unified Facilities Criteria

USACE—US Army Corps of Engineers

USAF—United States Air Force

WBDG—Whole Building Design Guide

Office Symbols

AF/A4—DCS/Logistics, Engineering & Force Protection
AF/A4C—Director of Civil Engineers

Terms

Base—A locality from which operations are projected or supported, or an area or locality containing installations which provide logistic or other support.

Basic Expeditionary Airfield Resources—Facilities, equipment, and basic infrastructure to support the beddown of deployed forces and aircraft at austere locations; a critical capability to fielding expeditionary aerospace forces. Also known as BEAR, the resources include tents, field kitchens, latrine systems, shop equipment, electrical and power systems, runway systems, aircraft shelters, and water distribution systems needed to sustain operations.

Beddown—The act of providing facilities, utilities, services, construction, operations and maintenance support to a deployed force with the overall intent of establishing a basic mission capability.

Contingency—A situation requiring military operations in response to natural disasters, terrorists, subversives, or as otherwise directed by appropriate authority to protect US interests.

Contingency Location—A non-enduring location outside of the United States that supports and sustains operations during named and unnamed contingencies or other operations as directed by appropriate authority and is categorized by mission life-cycle requirements as initial, temporary, or semi-permanent.

Force Beddown—The provision of expedient facilities for troop support to provide a platform for the projection of force.

Freeboard—The vertical distance between the water surface at the design flow (maximum expected water level) and the top edge of the ditch bank or embankment. It acts as a safety margin to prevent overflow due to unexpected surges, waves, or debris buildup.

Open-Air Burn Pit— An area that is designated for disposing of solid waste by burning in the outdoor air at a contingency.

Overseas—A geographic area located outside the jurisdiction of the United States, which includes land and associated territorial sea, contiguous zones, and exclusive economic zones of the United States; an area outside the United States (e.g., a foreign country).

Attachment 2

ENGINEER REACHBACK AND OTHER USEFUL LINKS

Table A2.1. Useful Organizational and Product Links.

Organization and Products Links
Air Force Civil Engineer Center: https://www.afcec.af.mil/
CE DASH (AFCEC Technical Support Portal): https://usaf.dps.mil/teams/CEDASH/scripts/homepage/home.aspx
CE Playbooks: https://www.ceplaybooks.com.
Air Force Publications and Forms: https://www.e-publishing.af.mil/
My Learning (Learning Management System): https://lms-jets.cce.af.mil/moodle/login/index.php
Air Force Design Guides: https://www.wbdg.org/ffc/af-afcec
Whole Building Design Guide: https://www.wbdg.org/
US Army Corp of Engineers Official Publications, http://www.publications.usace.army.mil/Home.aspx
Unified Facilities Criteria: https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc
Unified Facilities Guide Specifications: https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs
USACE Reach-back Operations Center: https://uroc.usace.army.mil
Army Publications and Forms: https://armypubs.army.mil/